The Intensive and Extensive Margins of the Decline in US Manufacturing*

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Abstract

We show that the decline of U.S. manufacturing is driven by the extensive margin: young workers disproportionately avoid entry into declining sectors, while incumbents exit through retirement. We document that the labor supply elasticity with respect to wages are 7 times higher for new entrants than for incumbents. We develop a model that shows this high-elasticity entrant margin creates quantity amplification and wage attenuation in the labor market in response to sectoral shocks, and a smaller underlying demand shock is sufficient to explain the observed decline compared to models lacking this margin. A multi-region quantitative application of our framework replicates key empirical facts, including local-level recoveries driven by young workers entering service sectors. Our results suggest adjustment policies may be more effective when targeting the high-elasticity extensive margin of new entrants rather than the low-elasticity intensive margin of incumbents.

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1 Introduction

U.S. manufacturing has been in retreat for generations. Between 2000 and 2017, the sector shed roughly six million jobs, even as real output held steady. A vast literature links that contraction to a combination of import competition, especially the "China shock" (Autor et al., 2013, 2021), rapid automation (Acemoglu and Restrepo, 2020), and other structural factors (Charles et al., 2019). But does this adjustment come mainly from incumbent workers switching out of declining sectors—the intensive margin—or from retirement of experienced workers and new entrants choosing not to step onto the shop-floor career ladder—the extensive margin?

The distinction between these two margins is important not only to describe worker flows per se, but because it helps identify the *elasticity of sectoral reallocation* that governs the economy's response to shocks and policy. If manufacturing contracts mainly through incumbents switching sectors, then the key object is the reallocation elasticity of experienced workers, and counterfactuals, such as the employment and wage effects of tariffs or industrial policy, depend on how readily those workers move when relative wages change.

If instead adjustment occurs primarily through young workers choosing different first jobs and older workers retiring, then aggregate adjustment is driven by the elasticity of entry and initial sector choice. These alternative elasticities imply different general-equilibrium paths for sectoral wages and employment since they jointly adjust. This equilibrium adjustment will also change which policies are most effective, from retraining and adjustment assistance targeted at incumbents to interventions that directly affect job creation and entry.

In this paper, we study the extensive margin in the decline of U.S. manufacturing by combining new stylized facts with a life-cycle model of sectoral choice. We begin with an accounting decomposition of the decline in the manufacturing employment share between 1980 and 2019 and find that approximately 80% of this decline is due to the extensive margin, with older workers exiting manufacturing jobs and a corresponding lack of entry by young workers. We then quantify the elasticities of adjustment and find that young entrants are far more responsive to sectoral wage differentials than old incumbents, with an elasticity 7 times larger.

This heterogeneity changes equilibrium adjustment: negative shocks to manufacturing generate smaller wage movements but larger manufacturing employment responses than models that treat mobility as uniform across workers. In a multi-region quantitative exercise, incorporating the entry margin reduces the size of the inferred manufacturing demand shock needed to match observed import exposure by 15.7 %.

A large literature following Artuç et al. (2010) estimates the dispersion of idiosyncratic preference shocks governing worker reallocation across sectors. This elasticity is a key parameter governing equilibrium labor market adjustment and welfare (Caliendo et al., 2019; Dvorkin, 2025; Seo and Oh, 2025), yet existing estimates are identified exclusively from incumbent workers' sector-switching behavior and then applied uniformly to all workers, including new en-

trants.

We present a dynamic discrete-choice model of sector choice with overlapping generations that allow incumbents and entrants to differ in their mobility elasticity. While heterogeneity in reallocation costs sets baseline mobility levels, it is the heterogeneity in elasticities that governs marginal responses to counterfactual shocks and policy experiments. Our estimation separately identifies these two objects: in our preferred specification, incumbents respond to a 1% decrease in wages by leaving 0.38 % more, while entrants respond by entering 2.71 % less—a ratio of approximately 7 that is stable across data sources, estimation methods, and assumptions about the cost structure.

In general equilibrium, we derive two theoretical implications of life-cycle heterogeneity in mobility elasticities. First, when entry and retirement decisions are substantially more elastic than incumbent reallocation, the aggregate sectoral labor supply becomes more responsive to shocks even if incumbent switching is limited. As a result, negative sectoral demand shocks generate wage attenuation and quantity amplification: relative wages move less, while employment adjusts more, than in models that impose a uniform wage elasticity. This creates a calibration wedge: omitting the entrant margin forces the model to attribute a given observed employment response to a larger underlying shift in labor demand.

Second, the composition of adjustment is inherently nonlinear. For small shocks, high-elasticity entrants absorb a disproportionate share of the marginal response, limiting the need for costly incumbent reallocation; as shocks grow, this "entrant slack" is gradually exhausted and additional adjustment increasingly falls on the low-elasticity incumbent margin. Taken together, these results imply a systematic bias in canonical models with a single mobility elasticity inferred from incumbent mobility: they overstate wage adjustment, understate employment adjustment, and—when calibrated to match quantities—overstate the magnitude of underlying demand shocks and the welfare costs borne by affected workers.

Our quantitative simulations in Section 5 discipline the mechanism with data and quantify its general-equilibrium and distributional implications in a multi-region setting. The model reproduces the key empirical finding that manufacturing adjustment is driven primarily by cohort replacement, and that employment recoveries in trade-exposed places occur mainly through new cohorts entering low-wage service jobs. Quantitatively, the entrant margin acts as a local buffer in high-manufacturing states, muting manufacturing wage losses by shifting adjustment toward quantities. It also generates a sizable calibration wedge: relative to a canonical model without entry, matching the same import-exposure facts requires a 15.7 % smaller underlying trade shock.

These results imply that policies targeting incumbent reallocation are unlikely to generate large sectoral transitions. Programs such as Trade Adjustment Assistance (TAA) may therefore have limited effects on aggregate reallocation, whereas interventions that operate through entry and local job creation—such as place-based incentives—are more likely to shape both the

speed of recovery and the incidence of shocks.

Related Literature

Recent micro-evidence supports our emphasis on the extensive margin. Autor et al. (2025b) show that commuting zones that lose manufacturing jobs later regain employment largely through young workers taking service-sector jobs, while incumbent manufacturing workers rarely switch sectors and suffer persistent earnings losses. We provide a structural interpretation of this pattern by estimating a large entrant–incumbent elasticity gap and tracing its implications for equilibrium adjustment and counterfactual policy analysis. This mechanism helps reconcile the continued decline of manufacturing employment after the China shock (Autor et al., 2021) with cohort-driven local recoveries and limited incumbent reallocation (Autor et al., 2025b).

Our paper contributes to the literature that embeds forward-looking labor supply decisions into quantitative trade models, pioneered by Artuç et al. (2010). This framework has been used to to study the surge in Chinese import competition (Caliendo et al., 2019; Dix-Carneiro et al., 2023; Kim et al., 2024; Rodriguez-Clare et al., 2025), tariff reforms and trade policy (Caliendo and Parro, 2022), place-based policies (Donald et al., 2025a; O'Connor, 2025), capital accumulation (Kleinman et al., 2023), refugee inflows (Brinatti and Guo, 2024; Caliendo et al., 2023), and environmental impact (Balboni, 2025). In these models, the elasticity of worker mobility to future wages is a crucial parameter governing the magnitude of equilibrium adjustments and welfare counterfactuals. We show that this key elasticity is sharply dependent on whether the agent is entering (extensive) or reallocating (intensive). Recognizing this heterogeneity materially changes the counterfactual predictions regarding the speed of labor market adjustment, the effect on wages, and the distributional consequences of trade shocks.

More broadly, our framework relates to a set of empirical findings in the trade and labor literature that emphasize persistent sectoral decline, cohort-based adjustment, and limited net reallocation despite substantial worker churn. By explicitly modeling the heterogeneous sectoral choices of new cohorts, we provide a unified structural interpretation of three patterns: (i) the continued decline of manufacturing employment after the China shock plateaued (Autor et al., 2021), (ii) a fall in manufacturing employment concentrated among younger cohorts and recovery through service ojbs (Autor et al., 2025b), and (iii) large gross sectoral flows but small net reallocation (Artuç et al., 2010). While our model highlights cohort-specific mobility, we abstract from other life-cycle dynamics, such as endogenous human capital accumulation, that may also shape long-run structural transformation.

Our findings also speak to the broader literature on long-run structural transformation, which analyzes the secular decline of agricultural and manufacturing employment (Pierce and Schott, 2016; Adão et al., 2024). Recent work such as (Porzio et al., 2022; Budı-Ors, 2025) show

that the decline in agriculture was also driven primarily by demographic turnover rather than incumbent reallocation. Cosar (2013); Hobijn et al. (2018) reach analogous conclusions for more general structural reallocation. We extend this line of inquiry in two ways. First, we provide a quantitative estimate of the cohort gap in mobility elasticities required to match recent U.S. data. Second, we trace the specific implications of this cohort-based adjustment mechanism for the incidence of trade shocks, thereby bridging the structural transformation and international trade literature.

2 Stylized Facts on the Margins of Manufacturing Decline

This section motivates our framework by documenting *how* U.S. manufacturing shrank: through incumbent reallocation across sectors (the intensive margin) or through cohort turnover—older workers leaving and young workers not entering (the extensive margin). Using linked CPS microdata, we decompose year-by-year changes in manufacturing employment into these margins and find that roughly four-fifths of the long-run decline is accounted for by the extensive margin. Moreover, the extensive margin is strongly life-cycle patterned: exits are concentrated among older workers (consistent with retirement), while inflows from young entrants steadily fall. This is precisely the margin that will govern labor-supply responsiveness in our model, because first-job sector choice is much more wage-elastic than sector switching by incumbents.

We proceed in three steps. First, we describe the linked CPS panel and our state classification. Second, we summarize what the raw transition matrices imply about life-cycle mobility. Third, we present three stylized facts that connect manufacturing's decline in shares and levels to (i) a post-2000 headcount break, (ii) a decomposition dominated by the extensive margin, and (iii) the central role of generational churn.

2.1 Data and Measurement

We use CPS Basic Monthly microdata from IPUMS (Flood et al., 2024), 1982–2019, and construct one-year links using the CPS month-in-sample structure (canonical MIS 4→16 match) and standard validation rules (consistent sex, race, quarter-of-birth/age, and state), dropping ambiguous matches; details are in Appendix B.1. Earnings are taken from the outgoing rotation groups and linked to subsequent one-year transitions using the same structure. The resulting linked sample contains large annual panels (90,000 matched individuals per year; Appendix B.1 reports match rates and sample sizes), which is well-suited to estimating aggregate transition rates and their time-series evolution. We focus on men aged 20–64 to avoid conflating manufacturing reallocation with secular changes in women's labor-force participation; results including women are reported in the Appendix. All moments are computed using CPS survey weights.

We classify each individual into one of four mutually exclusive states: manufacturing (M), agriculture (A), service (S), and non-employment (N). Unemployed individuals who report a prior industry are treated as temporarily separated and remain attached to that sector; thus, switches between M and $\{A, S\}$ are intensive-margin reallocations, while movements between M and N are extensive-margin transitions.[1] To isolate life-cycle entry and exit, we report results by age at t: young (20–29), prime-age (30–54), and old (55–64).

Table 1: Gross flows between four sector groups by age group. Means with standard deviations in parentheses.

	Nonemployment	Agriculture	Manufacturing	Service
Panel A. Young (20-29)				
Nonemployment	0.838	0.007	0.020	0.136
	(0.028)	(0.002)	(0.009)	(0.022)
Agriculture	0.231	0.642	0.023	0.104
	(0.039)	(0.033)	(0.009)	(0.024)
Manufacturing	0.135	0.005	0.759	0.101
	(0.034)	(0.003)	(0.029)	(0.025)
Service	0.165	0.004	0.020	0.810
	(0.023)	(0.001)	(0.006)	(0.022)
Panel B. Prime-age (30-54)				
Nonemployment	0.915	0.005	0.012	0.068
	(0.010)	(0.002)	(0.003)	(0.008)
Agriculture	0.148	0.774	0.016	0.062
	(0.023)	(0.021)	(0.006)	(0.017)
Manufacturing	0.080	0.003	0.847	0.069
	(0.019)	(0.002)	(0.021)	(0.024)
Service	0.091	0.003	0.015	0.891
	(0.014)	(0.001)	(0.004)	(0.012)
Panel C. Old (55-64)				
Nonemployment	0.978	0.003	0.002	0.017
	(0.006)	(0.002)	(0.001)	(0.005)
Agriculture	0.193	0.753	0.012	0.043
	(0.030)	(0.030)	(0.009)	(0.021)
Manufacturing	0.147	0.003	0.794	0.057
	(0.032)	(0.002)	(0.023)	(0.024)
Service	0.154	0.002	0.012	0.832
	(0.027)	(0.001)	(0.005)	(0.025)

Table 1 reports average one-year transition matrices by age group. Two patterns summa-

[1] Across our sample, 97% of unemployed workers report an industry, consistent with temporary separation or continued attachment, whereas roughly 95% of those out of the labor force report no industry.

rize the life-cycle structure that motivates our decomposition. First, mobility is front-loaded: young workers are the most likely to move across sectors and between employment and non-employment. Second, non-employment becomes close to absorbing near retirement: older workers who leave employment rarely re-enter, and older workers in manufacturing are much more likely to transition to N than to switch into another sector. These gross flows already suggest that manufacturing can contract primarily through exits to N even if job-to-job switching across sectors is limited; the next subsection formalizes this intuition using net-flow decompositions.

2.2 Flows and accounting decomposition

Let L_t^i denote the number of workers in state $i \in \{M, A, S, N\}$ at time t, and let F_t^{ij} denote the number of individuals transitioning from i to j between t and $t+1$. To accommodate entry to and exit from the CPS universe (and linkage mismatch), we add an outside state R. Then the change in sectoral employment levels admits the exact decomposition

$$L_{t+1}^i - L_t^i = \underbrace{\sum_{j \in \{M,A,S\}} \left(F_t^{ji} - F_t^{ij}\right)}_{\text{intensive (between employed sectors)}} + \underbrace{\left(F_t^{Ni} - F_t^{iN}\right)}_{\text{extensive (employment} \leftrightarrow \text{non-employment)}} + \underbrace{\left(F_t^{Ri} - F_t^{iR}\right)}_{\text{outside / residual}} . \tag{1}$$

The first term captures net job-to-job reallocation across employed sectors; the second captures net flows between the sector and non-employment; the residual absorbs entry/exit from the 20–64 CPS sample and any remaining stock-flow inconsistency.

We report results both in headcounts and employment shares. The corresponding share decomposition (and the age-aggregation identity used to separate within-age flows from between-age composition) are provided in Appendix B.1. We also report age-specific versions of (1); in the life-cycle interpretation, we summarize "exit" as $F_t^{MN}(\text{old})$ and "entry" as $F_t^{NM}(\text{young})$.

Because linked CPS flows need not exactly sum to observed origin and destination stocks, we reconcile the flow matrix via biproportional fitting so that row sums match L_t and column sums match L_{t+1}, leaving the R margins free. Appendix B.1 describes this procedure and additional data-construction details.

2.3 Stylized Facts

We document three facts that highlight the role of the extensive margin in the decline of manufacturing, with a focus on the China shock in the early 2000s. Before turning to the decompositions, it is useful to distinguish manufacturing employment in *shares* versus *levels*. A sector's employment share can fall even if its headcount is stable when aggregate employment is growing. By contrast, a sustained decline in headcount requires net outflows not offset by inflows,

making entry and exit from employment central objects for our model.

Fact 1. *Manufacturing's share of total employment falls steadily, but its headcount breaks sharply after 2000.*

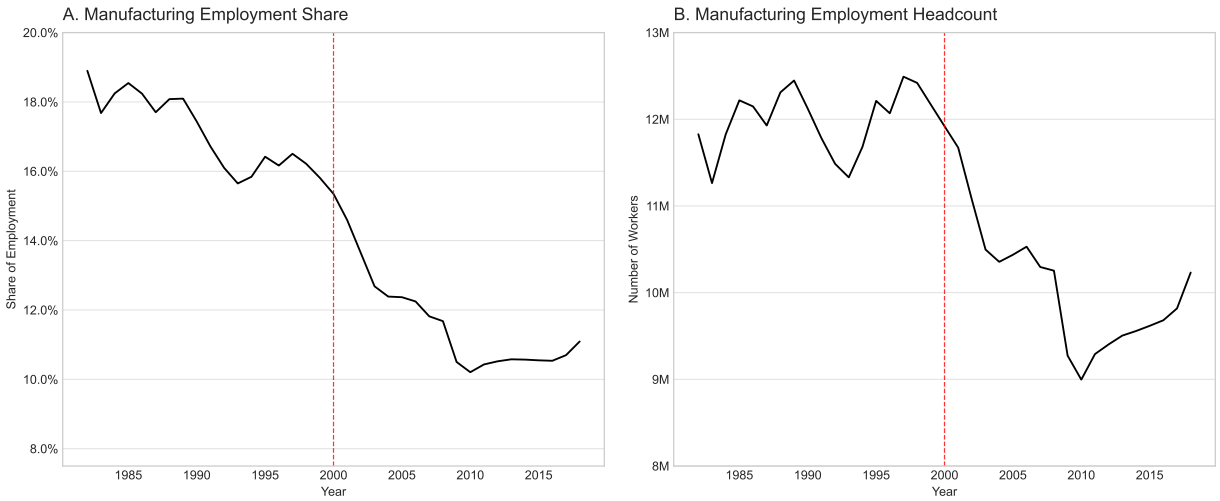


Figure 1: Divergence of Manufacturing Employment Share and Headcount.

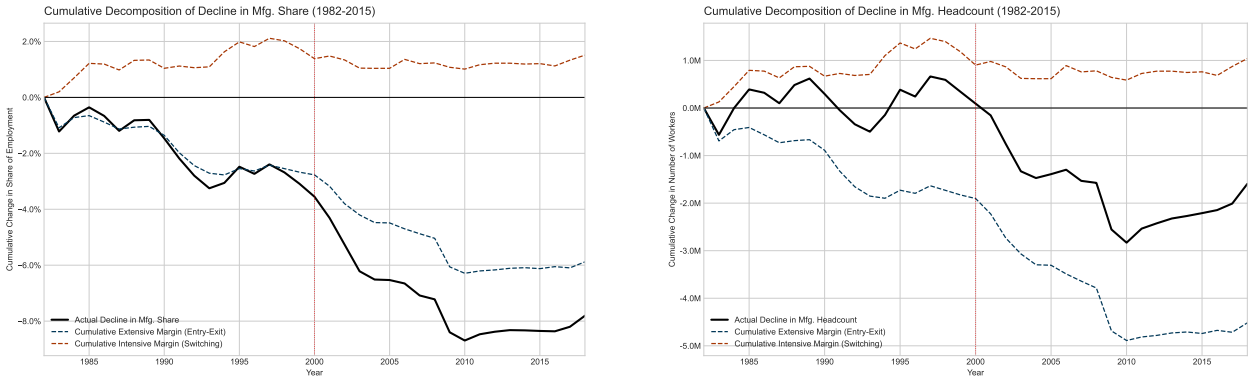
Note. This figure is computed from the linked CPS sample to align with the flow-based decomposition. Appendix B.1 reproduces the same series using BLS employment totals and shows the same divergence.

Figure 1 shows a key divergence: manufacturing's employment *share* declines smoothly throughout the sample, while manufacturing *headcount* is roughly flat through the late 1990s and then drops sharply after 2000. A falling share with flat headcount can arise mechanically from faster employment growth elsewhere; the post-2000 level break cannot. It requires persistent net losses of manufacturing workers—outflows to non-employment that are no longer offset by inflows. This distinction matters for interpreting the China shock: although the long-run share decline predates the 2000s, the discrete headcount contraction begins when import penetration accelerates, pointing to a deterioration in the extensive margin that we quantify in Facts 2–3.

Fact 2. *The decline in manufacturing is dominated by the extensive margin.*

Figure 2 decomposes annual changes in manufacturing employment into net job-to-job flows between employed sectors (intensive) and net flows between manufacturing and non-employment (extensive), using Equation (1) (and the share analogue in Appendix B.2). In both shares and levels, the intensive component is small relative to the overall decline and is close to zero on average; it is even mildly positive in the 1980s and 1990s. This is not inconsistent with deindustrialization: manufacturing can lose headcount primarily through exits to non-employment even if net switching *between employed sectors* is limited or positive.

Instead, the manufacturing contraction is driven by the extensive margin. After 2000, net ex-



(a) Change in share

(b) Change in headcount

Figure 2: Extensive vs Intensive Margin of Manufacturing Decline.

Note.

its from manufacturing to non-employment rise sharply and are not offset by inflows from non-employment into manufacturing, accounting for the bulk of the cumulative decline in headcount. The next fact shows that this extensive-margin adjustment reflects life-cycle turnover—older exit and missing young entry—rather than temporary non-employment during sector transitions in periods of deindustrialization.

Fact 3. *The extensive margin is driven by generational churn: high exit by older workers and low entry by young workers.*

Figure 3 disaggregates the net-flow decomposition by age at t and highlights the life-cycle structure behind the aggregate patterns. Panel A shows that net job-to-job switching (the intensive margin) is small for all age groups: the series is noisy and centered near zero, indicating that direct reallocation across sectors is not the main channel through which manufacturing shrinks.

The action is in the extensive margin. Panel B shows that exits from manufacturing to non-employment are consistently much larger for older workers (55–64) than for prime-age or young workers, consistent with retirement-like transitions. Panel C shows that these exits are not offset by corresponding inflows among young workers (20–29): net entry from non-employment into manufacturing is small and trends downward over time.

Panel D provides an explanation from the entry side. It plots manufacturing's share among all new job entrants for each age group and shows a secular decline in the sector's attractiveness to young cohorts—from about 20% in the early 1980s to below 10% by the 2000s. Taken together, the age patterns imply that manufacturing contracts primarily through cohort turnover—older workers leaving and younger workers increasingly starting elsewhere—rather than through incumbent sector switching.

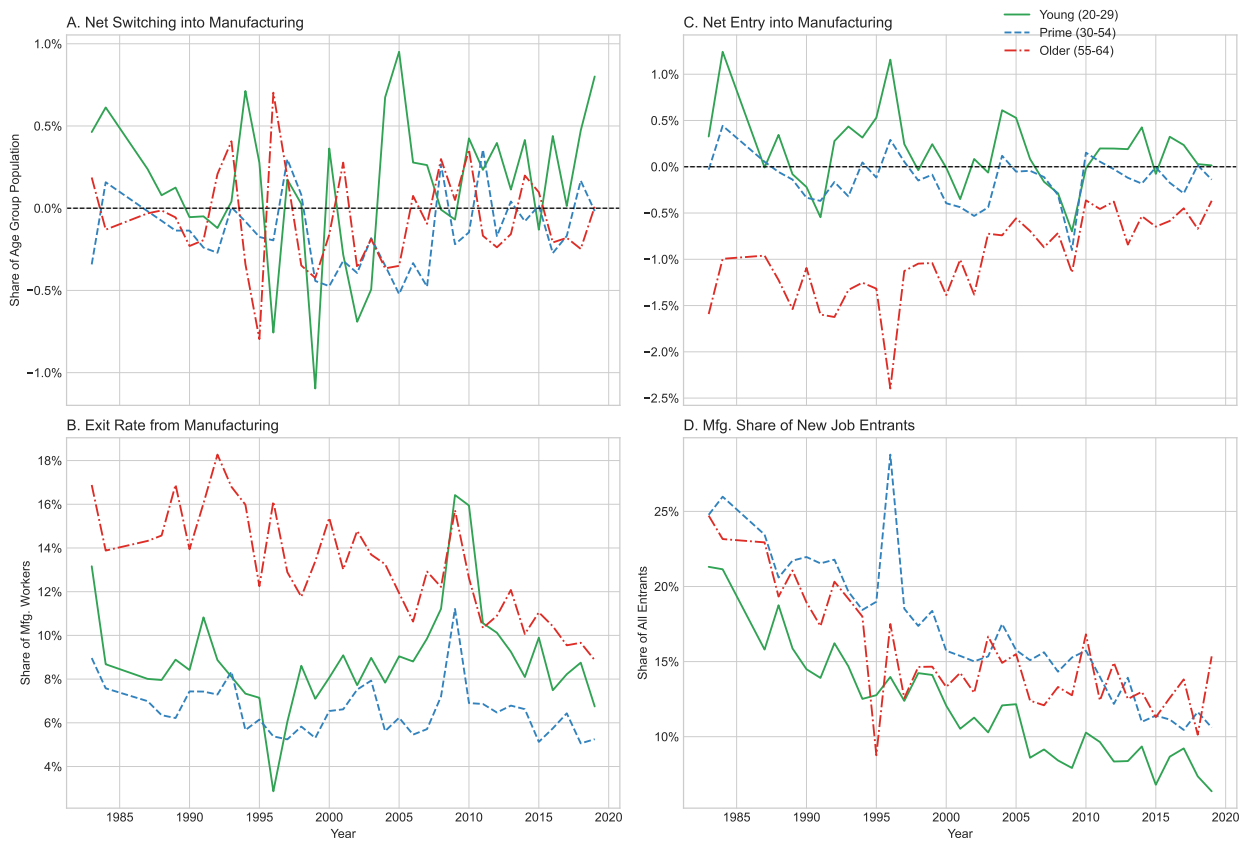


Figure 3: Extensive margin driven by old exit and lack of young entry.

2.4 Discussions

Summary. Taken together, Facts 1–3 imply that manufacturing's long-run contraction is primarily a cohort-turnover process: older workers exit manufacturing into non-employment and are not replaced by young entrants, while net sector switching among incumbents plays a limited role. This life-cycle structure motivates our model, which distinguishes entrants and incumbents and allows their mobility elasticities to differ.

Relation to prior evidence. These national flow patterns echo and extend the commuting-zone evidence in Autor et al. (2025b). In trade-exposed places, they show that manufacturing employment falls, total employment later recovers in services, and incumbent reallocation explains only a small share of the trade-induced decline, with recovery driven largely by young and foreign-born entrants. Our accounting decomposition shows that the same mechanism—weak incumbent switching and adjustment through cohort replacement—is also visible in linked national microdata and over the broader secular decline, not only in China-shock-exposed local labor markets.

Robustness. We externally validate the CPS-based measures using two alternative data sources.

First, we replicate the key entry and margin decompositions in administrative LEHD Job-to-Job (J2J) data from 2001 onward. The CPS object most central to our mechanism—young inflows into manufacturing from non-employment—maps naturally to LEHD hires from persistent non-employment (`NEPersist`), and the resulting entry-share series closely tracks the CPS measure and exhibits the same secular decline. LEHD job-to-job flows (`EE`) further sharpen the interpretation: switching into manufacturing is flat or rising for young workers, while the contraction occurs through the extensive inflow channel (Appendix B.3). Second, as a check that does not rely on month-in-sample linking, we replicate the decomposition using the CPS ASEC ("March CPS"), which records both current industry and the industry worked most in the previous calendar year; results are similar (Appendix **??**).

Taken together, the evidence points to a central role for the extensive margin of entry and retirement. In the next section, we develop a model with entrants and incumbents that captures this life-cycle structure and allows us to study mobility elasticities, welfare, and policy in transition.

3 A Model of Labor Supply with Entry and Exit

Section 2 highlights a set of facts about adjustment to sectoral decline. Manufacturing employment falls primarily through cohort turnover rather than incumbent sector switching: older workers exit to non-employment, and young cohorts increasingly do not enter manufacturing, even though gross worker churn is substantial. This suggests that the relevant object for understanding aggregate adjustment is not the mobility of the average incumbent, but the *elasticity of the entry margin*—how strongly new cohorts respond to expected wage paths when choosing their first sector.

Our goal in this section is to build a tractable labor-supply block that (i) cleanly separates the intensive margin (incumbent sector switching) from the extensive margin (entry and retirement), (ii) allows these margins to have different wage elasticities, and (iii) delivers quantitative implications for wage pass-through, employment adjustment, and policy counterfactuals in general equilibrium. The model provides a structural interpretation of the cohort-driven local recoveries and our national flow decompositions, and it clarifies what these facts imply for equilibrium adjustment and the calibration of sectoral shocks.

We first present the worker problem for incumbents and entrants and the law of motion for sectoral employment. We then show how the model maps observed flows and wages into two mobility elasticities, each for incubents and entrants, which we estimate in Section 3.2. Section 4 derives the key equilibrium implications: with an elastic entry margin, sectoral shocks generate wage attenuation and employment amplification, and models that omit entry require larger implied demand shocks to match the same quantity responses.

Environment. We adapt the dynamic discrete-choice framework of Artuç et al. (2010) to a perpetual youth overlapping-generations setting as in Blanchard (1985). Time is discrete. Workers choose among S productive sectors $s \in \{1, \ldots, S\}$ and a non-employment state $s = 0$. Each period, a mass N_t of new entrants arrives and chooses an initial sector. Incumbents earn sectoral wages $\{w_t^s\}$, can switch sectors subject to mobility costs, and retire permanently with probability $1 - \theta$.[2]

3.1 Incumbents and Entrants

Incumbents. An incumbent employed in sector s at the beginning of period t earns wage w_t^s and chooses sector n for period $t+1$. Moving from s to n entails a utility cost χ^{sn} (with $\chi^{ss} = 0$) and an idiosyncratic taste shock ϵ_t^n. Incumbents retire with probability $1 - \theta$ each period. Let β denote the discount factor and let $V_{t+1}^n = \mathbb{E}[\mathcal{V}_{t+1}^n(\epsilon)]$ denote the ex-ante value of starting period $t+1$ in sector n. The realized lifetime utility for an incumbent in sector s is

$$\mathcal{V}_t^s(\epsilon) = u(w_t^s) + \max_n [\beta\theta V_{t+1}^n - \chi^{sn} + \epsilon_t^n]$$

Assuming ϵ_t^n is i.i.d. Type I Extreme Value with dispersion parameter $\nu > 0$, the probability that an incumbent in sector s chooses sector n is

$$\mu_t^{sn} = \frac{\exp\left(\frac{1}{\nu}\left(\beta\theta V_{t+1}^n - \chi^{sn}\right)\right)}{\sum_{n'} \exp\left(\frac{1}{\nu}\left(\beta\theta V_{t+1}^{n'} - \chi^{sn'}\right)\right)} \tag{2}$$

and the ex-ante value function V_t^s satisfies the following equation:

$$V_t^s = u(w_t^s) + \nu \log\left(\sum_{n'} \exp\left(\frac{1}{\nu}\left(\beta\theta V_{t+1}^{n'} - \chi^{sn'}\right)\right)\right) \tag{3}$$

Entrants. An entrant at the end of period t chooses an initial productive sector $n \in \{1, \ldots, S\}$. Choosing n yields the discounted continuation value of becoming an incumbent in that sector next period, net of an entry cost C^n, plus a taste shock $\epsilon_{t,ent}^n$:

$$\mathcal{V}_t^{ent}(\epsilon) = \max_n \left[\beta V_{t+1}^n - C^n + \epsilon_{t,ent}^n\right]$$

Assuming $\epsilon_{t,ent}^n$ is Type I Extreme Value with dispersion parameter ν^E, the share of entrants choosing sector n is

$$E_t^n = \frac{\exp\left(\frac{1}{\nu^E}\left(\beta V_{t+1}^n - C^n\right)\right)}{\sum_k \exp\left(\frac{1}{\nu^E}\left(\beta V_{t+1}^k - C^k\right)\right)} \tag{4}$$

[2] We consider extensions with richer life-cycle dynamics in the Appendix.

Sectoral Labor Supply. Let L_t^s be the number of old workers in sector s at time t, and let N_t be the mass of new entrants. Applying the law of large numbers for a continuum of workers, the distribution of workers evolves according to:

$$L_{t+1}^n = \sum_s \mu_t^{sn} \theta L_t^s + E_t^n N_t \tag{5}$$

Equation (5) nests the workhorse model without cohort replacement as the special case $\theta = 1$ and $N_t = 0$. In the data, μ_t^{sn} corresponds to intensive-margin job-to-job transitions between employed sectors, while E_t^n corresponds to the entry margin proxied by young inflows from non-employment (Section 2).

Interpretation and use. Equations (2)–(5) fully characterize the labor-supply side of the model: given a path of sectoral wages $\{w_t^s\}$, they determine sectoral values $\{V_t^s\}$, transition probabilities $\{\mu_t^{sn}\}$, entrant shares $\{E_t^n\}$, and sectoral employment dynamics $\{L_t^s\}$. The key sufficient statistics are the two dispersion parameters ν and ν^E, which govern the wage responsiveness of incumbent switching and entry. Because this labor block can be combined with any demand system that maps $\{L_t^s\}$ into $\{w_t^s\}$, it can be embedded in quantitative models of trade or structural transformation to study equilibrium transition paths and counterfactual policies, as we do in Section 5. Next, we estimate ν and ν^E in Section 3.2.

3.2 Estimating entrant and incumbent mobility elasticities

The parameters ν, ν^E govern labor supply dynamics, and are central to welfare and counterfactual analysis. We adapt the approach of Artuç et al. (2010), which derives moment conditions from the agent's optimality conditions.

Moment Conditions. By manipulating the logit equations, we can derive a pair of Euler-like equations that relate current entry and reallocation decisions μ_t, E_t to expected future wages and future migration flows as continuation values. For incumbents, the choice between staying in sector s and moving to sector n yields the following relationship:

$$\log(\mu_t^{sn}/\mu_t^{ss}) = \frac{\beta\theta}{\nu}[u(w_{t+1}^n) - u(w_{t+1}^s)] + \beta\theta \log(\mu_{t+1}^{sn}/\mu_{t+1}^{nn}) - \frac{1-\beta\theta}{\nu}\chi^{sn} + \Omega_{t+1}^{sn} \tag{6}$$

For entrants, the decision to join sector n is characterized by:

$$\log E_t^n = \frac{\beta}{\nu^E} u(w_{t+1}^n) + \beta\theta \log(E_{t+1}^n) - \beta\theta \log \mu_{t+1}^{nn} + \alpha_t - \frac{1-\beta}{\nu^E} C^n + \Omega_{t+1}^{n,E} \tag{7}$$

The derivations are provided in Appendix A.1. The moment conditions have a recursive structure, as in Artuç et al. (2010). Equation (6) shows that an incumbent's relative probability of moving from s to n today depends on the expected future wage gain and the relative continuation value of being in sector n, which is proxied by future relative mobility. Similarly, Equation (7) links the entry choice today to the next period's wage and the long-run value of joining the sector, proxied by future entry E_{t+1} and incumbent retention rates μ^{nn}_{t+1}.

Notice that the coefficient on the retention rate $\log \mu^{nn}_{t+1}$ in Equation (7) is $\beta\theta$. As we show in Appendix A.1, an unconstrained derivation yields a factor of $\frac{\nu}{\nu^E}$, where ν is the average elasticity of the entire incumbent pool. However, the continuation value for a marginal entrant is more accurately captured by the elasticity of a first-year incumbent. We impose the restriction that this early-tenure elasticity equals the entrant elasticity ($\nu^{\text{1-year incumbent}} = \nu^E$), which yields $\beta\theta$. This naturally aligns with the empirical reality of labor market transitions and avoids the instability of estimating an aggregate ratio for a marginal decision.[3]

The error terms Ω^{sn}_{t+1} and $\Omega^{n,E}_{t+1}$ are expectation errors and are orthogonal to information available at time t. However, shocks realized in $t+1$ affect both the realized future objects on the right-hand side (e.g., w_{t+1} and the continuation-value proxies μ_{t+1}, E_{t+1}) and the expectation errors, so OLS is biased. We therefore instrument the $t+1$ right-hand-side variables with predetermined information at time t, using lagged values of sectoral wages and flows (e.g., w_{t-1}, μ_{t-1}, E_{t-1}) and sector fixed effects, following the standard approach in Artuç et al. (2010). Appendix A.1 reports the exact instrument set and robustness to alternative lag structures.

Identification and Interpretation

Equations (6)-(7) give a natural interpretation of the parameters ν and ν^E as an elasticity: $\frac{1}{\nu} = \frac{1}{\beta}\frac{d\log\mu}{du(w)}$, $\frac{1}{\nu^E} = \frac{1}{\beta}\frac{d\log E}{du(w)}$. These elasticities are sufficient statistics for a range of policy counterfactuals and welfare calculations. For instance, in a steady state, the lifetime utility of an incumbent in sector s can be expressed as $V^s = \frac{1}{1-\beta\theta}(u(w^s) - \nu \log \mu^{ii})$, directly linking welfare to the parameter ν and the observable stayer share μ^{ii}.[4]

The estimation framework allows for the separate identification of sector-specific reallocation costs and labor supply elasticities. The average levels of labor flows identify the cost parameters relative to the dispersion parameter (i.e. χ^{sn}/ν and C^n/ν^E). In contrast, the elasticities themselves (ν and ν^E) are identified by how these flows respond over time to changes in wages. To identify costs that vary by origin and destination, we estimate a stacked version of the equations across all time periods, including origin-and-destination fixed effects. For comparability with prior literature, we also present estimates under more restrictive assumptions about the cost structure (e.g., constant or destination-specific costs).

[3] We discuss the full derivation and associated robustness exercises in Appendix A.1.

[4] With an appropriate demand system, the sufficient statistics for dynamic counterfactuals are ν, the observed reallocation flows, and trade flows (Caliendo et al., 2019; Rodriguez-Clare et al., 2025).

Based on our descriptive evidence in Section 2, we can form priors on the relative magnitudes of these parameters. We expect entrants, who have not yet accumulated sector-specific human capital, to be more responsive to wage signals than incumbents. A higher labor supply elasticity for entrants corresponds to a smaller dispersion parameter, leading to the testable hypothesis that $\nu^E < \nu$.

For estimation, we require annual sector-to-sector transition rates and sectoral wages. We construct one-year linked CPS panels using the month-in-sample structure (canonical MIS 4→16 match) as described in Section 2 and Appendix B.1. The data spans 1983–2019 and comprises 2,991,559 unique individuals (approximately 80,000 observations a year, weighted using standard CPS weights). While the four-state partition in Section 2 is used for the stylized-facts accounting, for estimation we match the model's choice set by mapping CPS industries into the six-sector aggregation of Artuç et al. (2010).[5]

Using this classification, we compute incumbent transition probabilities μ_t^{sn} from observed job-to-job reallocations. Sectoral wages w_t^s are constructed from outgoing rotation group (ORG) weekly earnings, deflated and cleaned as in Section 2. We proxy entry shares E_t^n using young workers' (ages 20–29) transitions from non-employment (N) into sector n, which captures school-to-work and re-entry flows in the absence of true first-job histories. Throughout we assume $u(w) = \log w$, so the estimated coefficients map directly into the wage elasticities $\beta\theta/\nu$ and β/ν^E.

Estimators. We report two estimators for the incumbent equation (6): an IV specification that instruments (w_{t+1}, μ_{t+1}) with $t-1$ lags and sector dummies, and a two-stage Poisson pseudo-maximum-likelihood (PPML) procedure following Artuç and McLaren (2015) that estimates the gravity-style flow equation in levels. The PPML approach identifies ν but does not separately identify the full set of bilateral cost terms χ^{ij}; we therefore report the implied ν and omit χ^{ij} in that panel (details in Appendix A.1).

Table 2 and Table 3 report estimates of the dispersion parameters that govern mobility: ν for incumbents and ν^E for entrants. With log utility $u(w) = \log w$, these parameters map directly into the wage responsiveness of flows. For incumbents, the coefficient on the expected wage differential in (6) is $\beta\theta/\nu$, so a smaller ν implies a larger responsiveness of relative switching probabilities to wages. For entrants, the coefficient on the expected wage level in (7) is β/ν^E, so a smaller ν^E implies a larger responsiveness of entry shares to wages.

Incumbent mobility (ν). Table 2 shows that incumbent mobility is relatively inelastic across specifications. Under our benchmark IV specification with dyadic switching costs (column (1)), we estimate $\nu = 2.44$, implying an incumbent wage elasticity of approximately $\varepsilon^{\text{inc}} \equiv \beta\theta/\nu \approx$

[5]The six sectors are: (1) agriculture and mining; (2) construction; (3) manufacturing; (4) transportation, communication and utilities; (5) trade; (6) other services.

Table 2: Incumbent mobility parameter ν

	(1)	(2)	(3)
χ specification	Dyadic (χ^{ij})	Entry–exit ($\bar{\chi}_i + \tilde{\chi}_j$)	Constant (χ)
Panel A. IV			
	2.44^{*}	2.22^{*}	1.71^{***}
	(1.86)	(1.48)	(0.11)
Panel B. Two-stage PPML			
	1.14^{***}	2.48^{***}	1.31^{***}
	(0.07)	(0.24)	(0.10)
Observations		1050	

Note: Standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$ (one-tailed). Switching cost estimates χ^{ij} reported in the appendix.

Table 3: Entry parameter ν^E

	(1)	(2)
C^i specification	C^i	$C^i = C$
ν^E	0.350^{***}	0.070^{***}
	(0.094)	(0.005)
Observations	175	

Note: Standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$ (one-tailed). Entry costs are identified only up to a normalization and are reported in the appendix.

0.38. Alternative cost restrictions (columns (2)–(3)) and the PPML estimator yield similar magnitudes, reinforcing the conclusion that incumbent sector switching responds weakly to wage differentials.

Entry (ν^E). Table 3 shows substantially higher responsiveness on the entry margin. The preferred IV specification with sector-specific entry costs (column (1)) yields $\nu^E = 0.350$, implying an entrant wage elasticity $\varepsilon^{\text{ent}} \equiv \beta/\nu^E \approx 2.71$. Imposing constant entry costs (column (2)) produces an even smaller ν^E, but we view sector-specific entry costs as more realistic and therefore use $\nu^E = 0.350$ as the benchmark.

The model-implied gap in responsiveness between the extensive and intensive margin is the ratio of these elasticities:

$$\frac{\varepsilon^{\text{ent}}}{\varepsilon^{\text{inc}}} = \frac{\beta/\nu^E}{\beta\theta/\nu} = \frac{\nu}{\theta\nu^E}.$$

Plugging in our benchmark IV estimates (and $\theta \simeq 0.97$) yields $\varepsilon^{\text{inc}} \approx 0.38$ and $\varepsilon^{\text{ent}} \approx 2.71$, so entrants respond about $2.71/0.38 \approx 7$ times more than incumbents. Interpreted locally, a 1% decline in the sectoral wage path reduces entrant inflows by about 2.71 % but raises incumbent outflows only by about 0.38 %.

This entrant–incumbent elasticity gap provides a structural interpretation of the "generational churn" patterns in Section 2. Manufacturing can keep shrinking even when incumbent switching is limited because steady outflows are generated from retirement, while forward-looking young cohorts disproportionately avoid entering the sector when expected wage paths deteriorate.

Robustness. Table 4 shows that the entrant–incumbent elasticity ratio $\nu/\nu^E \approx 7$ is stable across a wide range of specification choices, even when the underlying elasticities themselves are sensitive to data sources and parameter choices. Across both data sources, the ratio ranges from 4.2 to 9.7, with most estimates clustering between 6 and 7. Varying the IV lag from 2 to 3, the discount factor β from 0.90 to 0.97, or excluding recession years all leave the ratio largely unchanged. The individual parameters ν and ν^E do move with these choices — most notably, Monthly CPS and March CPS (ASEC) yield markedly different levels of ν (roughly 2 vs. 5) and ν^E (roughly 0.35 vs. 0.70) — but they move together, so their ratio remains close to 7.[6] This suggests that the entrant–incumbent gap is a robust feature of the data rather than an artifact of any particular modeling choice.

3.3 Discussions

Relation to Empirical Findings. Our finding that adjustment occurs primarily on the entry margin resonates with the commuting-zone level evidence in the literature. For example, it helps explain why manufacturing employment continues to decline even after trade shocks plateau around 2010, and exposed places ultimately recover overall employment via entrants into services while incumbent employment remain depressed and manufacturing continues to shrink, as documented by Autor et al. (2021) and Autor et al. (2025b). A likely mechanism for this dynamic is that incumbents are quite immobile due to their sector-specific human capital, and structural reallocation occurs mainly via new cohorts choosing service sector jobs. In the next section, we conduct counterfactual experiments both with and without entry to show how admitting the extensive margin changes the size of shocks needed to match import penetration

[6]This discrepancy partly reflects a timing bias documented by Kambourov and Manovskii (2013): although March CPS mobility appears annual, respondents likely report employment from a sub-annual window, causing flow rates and estimates of ν to be biased upward. Artuç et al. (2010) correct for this by annualizing the observed flow matrix as $\tilde{\mu}^{12/K}$, finding $K = 5$ months best reconciles March CPS flows with NLSY data. For our purposes, however, the ratio ν/ν^E remains stable across datasets and correction procedures—suggesting that while the level of ν is sensitive to timing assumptions, the relative dispersion is not.

Table 4: Robustness of entrant–incumbent elasticity ratio ν/ν^E

	ν	ν^E	ν/ν^E
Monthly CPS			
Baseline	2.44 (1.86)	0.35 (0.09)	7.02 (5.69)
Excl. recessions	2.59 (2.62)	0.61 (0.26)	4.22 (4.62)
IV lag = 3	1.77 (0.20)	0.27 (0.08)	6.45 (1.96)
$\beta = 0.90$	1.65 (0.89)	0.34 (0.09)	4.87 (2.94)
$\beta = 0.97$	2.97 (2.72)	0.35 (0.09)	8.47 (8.09)
March CPS (ASEC)			
Baseline	4.67 (2.77)	0.72 (0.23)	6.52 (4.40)
Excl. recessions	3.69 (1.99)	0.58 (0.16)	6.38 (3.84)
IV lag = 3	9.85 (3.84)	1.02 (0.53)	9.68 (6.33)
$\beta = 0.90$	4.26 (2.39)	0.70 (0.22)	6.11 (3.95)
$\beta = 0.97$	4.86 (2.97)	0.72 (0.23)	6.72 (4.64)
FE specification			
ν: Dyadic FE (Dest FE for IV lag = 3)			
ν^E: Sector FE			

Note: Standard errors in parentheses (delta method for ν/ν^E). Benchmark specification uses Monthly CPS with dyadic switching costs, sector-specific entry costs, IV lag = 2, and $\beta = 0.95$. Full results across all FE combinations are reported in the Appendix.

paths and the distribution of incidence across cohorts.

Non-employment. Our estimation focuses on mobility across the S productive sectors. Following Artuç et al. (2010), we model non-employment as an involuntary process rather than as a sector of choice for incumbents. This modeling choice is motivated by the empirical findings in Section 2, where flows into and out of non-employment exhibit distinct patterns from inter-sectoral switching, suggesting they are driven by different mechanisms. While this approach maintains a tractable choice set, we acknowledge it as a limitation, as it abstracts from potentially important dynamics at the extensive margin of employment.

Implications. The estimates in this section reduce the labor-supply side of the model to a small set of objects: baseline worker flows $\{\mu_t^{sn}, E_t^n, L_t^s\}$, which are directly observed, and two elasticity composites $\beta\theta/\nu$ and β/ν^E, which we have estimated. A natural question is whether these objects are sufficient for counterfactual analysis—or whether one also needs to recover the full matrix of switching costs χ^{sn} and entry costs C^n, which are high-dimensional and only partially identified. The next section shows that the answer is no: cost levels are irrelevant for counterfactual dynamics, and the estimated elasticities together with observed flows are all

that is needed.

4 The Elasticity Gap and Adjustment Dynamics

Section 3.2 delivered two key estimates: incumbents are relatively immobile ($\nu \approx 2.44$), while entrants respond roughly seven times more elastically to wage signals ($\nu/\theta\nu^E \approx 7$). This section develops the theoretical implications of this finding. We first show that the baseline flows $\{\mu_t^{sn}, E_t^n\}$ and the estimated elasticities are sufficient statistics for counterfactual dynamics—the underlying cost levels are not needed (Section 4.1). We then unpack what this means for equilibrium outcomes. Locally, the elastic entry margin amplifies quantity responses and attenuates wage pass-through (Section 4.2), with the burden of adjustment allocated across margins according to a slack-elasticity product (Section 4.3). For permanent shocks, the entry margin front-loads the transition path and reduces the persistence of employment deviations (Section 4.4).

To analyze general equilibrium effects, we augment the labor supply framework with a generic inverse labor demand function for each sector s:

$$w_t^s = f^s(L_t^s, \{\xi_t^n\}) \tag{8}$$

where L_t^s is the total labor employed in sector s at time t, and $\{\xi_t^n\}$ represents exogenous sector-specific demand shifters (e.g., technology, trade prices, preferences) across all sectors. We assume standard conditions for the demand function ($df^s/dL^s < 0$, $df^s/d\xi^s > 0$, $df^s/d\xi^n < 0$). The labor market equilibrium is determined by the intersection of labor supply (Equations 2-5) and labor demand (Equation 8).

4.1 Elasticities as Sufficient Statistics

Following the exact hat algebra approach of **?** and Caliendo et al. (2019), we express counterfactual dynamics entirely in changes relative to a baseline equilibrium. The key observation is that baseline transition probabilities and entry shares already encode all information about cost levels, so that counterfactual flows can be written as functions of these observable flows, changes in continuation values, and the two elasticity composites $\beta\theta/\nu$ and β/ν^E estimated in Section 3.2—which are therefore sufficient statistics for counterfactual analysis.

Consider a counterfactual economy that shares the primitives $(\beta, \theta, \nu, \nu^E)$ and choice set of the baseline but allows different wages $\tilde{w}_t^s$ and possibly different costs $\tilde{\chi}^{sn} = \chi^{sn} + \Delta\chi^{sn}$ and $\tilde{C}^n = C^n + \Delta C^n$. Let $\Delta V_t^s := \tilde{V}_t^s - V_t^s$. Since $u(w) = \log w$, define wage changes in hat notation

and policy wedges as

$$\hat{w}_t^s := \frac{\tilde{w}_t^s}{w_t^s}, \qquad \hat{\kappa}^{sn} := \exp\big(-\Delta\chi^{sn}/\nu\big), \qquad \hat{\kappa}_E^n := \exp\big(-\Delta C^n/\nu^E\big),$$

so that $\Delta u_t^s = \log \hat{w}_t^s$. We have the following:

Proposition 1 (Exact hat algebra for the entrant–incumbent block)**.** *Suppose $\{\log \hat{w}_t^s\}$ is bounded. Then the system admits a unique bounded solution $\{\Delta V_t^s\}$, and the counterfactual path satisfies:*

$$\Delta V_t^s = \log \hat{w}_t^s + \nu \log\left(\sum_n \mu_t^{sn}\, \hat{\kappa}^{sn} \exp\left(\tfrac{\beta\theta}{\nu}\Delta V_{t+1}^n\right)\right), \tag{9}$$

$$\tilde{\mu}_t^{sn} = \frac{\mu_t^{sn}\, \hat{\kappa}^{sn}\, \exp\left(\frac{\beta\theta}{\nu}\Delta V_{t+1}^n\right)}{\sum_k \mu_t^{sk}\, \hat{\kappa}^{sk}\, \exp\left(\frac{\beta\theta}{\nu}\Delta V_{t+1}^k\right)}, \tag{10}$$

$$\tilde{E}_t^n = \frac{E_t^n\, \hat{\kappa}_E^n\, \exp\left(\frac{\beta}{\nu^E}\Delta V_{t+1}^n\right)}{\sum_k E_t^k\, \hat{\kappa}_E^k\, \exp\left(\frac{\beta}{\nu^E}\Delta V_{t+1}^k\right)}, \tag{11}$$

$$\tilde{L}_{t+1}^n = \theta \sum_s \tilde{L}_t^s\, \tilde{\mu}_t^{sn} + \tilde{N}_t\, \tilde{E}_t^n, \tag{12}$$

where $\tilde{N}_t$ follows a population rule. The levels $\{\chi^{sn}, C^n\}$ matter only to the extent that they determine the baseline flows $\{\mu_t^{sn}, E_t^n\}$. Counterfactual dynamics depend on structural parameters through the elasticity composites $(\beta\theta/\nu,\ \beta/\nu^E)$; policy interventions that alter costs enter only through the wedges $\hat{\kappa}^{sn}$ and $\hat{\kappa}_E^n$.

Proposition 1 characterizes the labor-supply side of any counterfactual. Wages $\hat{w}_t^s$ are not determined by (9)–(12) alone: they depend on the labor demand system—firm technology, trade costs, world prices—which maps sectoral employment $\tilde{L}_t^s$ into wages through (8). The proposition's content is that *whatever* demand system one embeds, the labor-supply side of the counterfactual never requires cost levels $\{\chi^{sn}, C^n\}$. These are absorbed entirely by the baseline flows $\{\mu_t^{sn}, E_t^n\}$, so the only objects the supply block contributes to the joint system are the elasticity composites $\beta\theta/\nu$ and β/ν^E and, when costs are themselves the object of intervention, the policy wedges $\hat{\kappa}^{sn}$ and $\hat{\kappa}_E^n$.

Next, to build intuition for how these elasticities shape equilibrium outcomes, we derive a series of *local* results on counterfactual economies near the observed equilibrium.

4.2 Wage Pass-Through and Employment Adjustment

Section 2 documented that manufacturing employment declined primarily through cohort replacement rather than incumbent switching, and Section 3.2 showed that entrants are roughly

seven times more elastic than incumbents. What are the equilibrium implications of this gap? We now show that an elastic entry margin amplifies employment responses to demand shocks while attenuating wage responses, and that models omitting this margin require larger shocks to match observed employment paths.

To formalize this, consider a partial-equilibrium experiment: fix a sector s and trace the effect of a marginal change in sector-s wage w_s^{t+1} on next-period employment L_{t+1}^s, holding other wages and future valuations $\{V_{t'}^{s'}\}_{t' \geq t+2}$ fixed. As derived in the Appendix (and summarized in Lemma 1), under log utility $u(w) = \log w$, the resulting supply response aggregates the contributions from both margins:

Lemma 1 (Labor supply response to wages)**.** *The marginal response of labor supply to future wages is given by*

$$x_t^s := \frac{\partial L_{t+1}^s}{\partial w_{t+1}^s} = \frac{\beta}{w_{t+1}^s}[\underbrace{\frac{\theta^2}{\nu}\sum_n L_t^n \mu_t^{ns}(1-\mu_t^{ns})}_{\textit{incumbent margin}} + \underbrace{\frac{1}{\nu^E} E_t^s(1-E_t^s)N_t}_{\textit{entrant margin}}] > 0. \tag{13}$$

The total supply response depends on the respective mobility elasticities ($1/\nu$, $1/\nu^E$) and the "slack" in each margin, captured by $\mu_t^{ns}(1-\mu_t^{ns})$ for incumbents moving from n to s and $E_t^s(1-E_t^s)$ for entrants choosing s. Since $\nu^E \ll \nu$, the entry margin can dominate the overall supply response even if entrants are a modest share of the labor force, provided the entry probabilities are not near the boundary $0, 1$.

This heightened supply responsiveness has direct equilibrium consequences:

Proposition 2 (Entry, Wage Pass-Through, and Calibration Wedge)**.** *Consider a negative labor demand shock $\Delta\xi_{t+1}^s < 0$. Compared to a model without an active (E_t^s not near 0, 1) and more elastic entry margin ($\nu^E < \nu$),*

(a) Wage pass-through is attenuated: The equilibrium wage response $|\partial w_{t+1}^s / \partial \xi_{t+1}^s|$ is smaller.

(b) Quantity response is amplified: The equilibrium employment response $|\partial L_{t+1}^s / \partial \xi_{t+1}^s|$ is larger.

(c) Calibration implications: To match a given observed employment change ΔL_{t+1}^s, the required shock $|\Delta\xi_{t+1}^s|$ is smaller.

Proof. (Sketch, see Appendix for full proof) Because entrants are more responsive ($\nu^E < \nu$), a negative shock causes them to avoid the affected sector more strongly. This larger quantity adjustment (increase in x_t^s) dampens the fall in the equilibrium wage needed to clear the market. Consequently, attributing an observed employment decline solely to incumbent responses (ignoring the more elastic entry margin) would lead to overestimating the magnitude of the underlying demand shock ξ_{t+1}^s. □

Proposition 2 highlights a key bias in quantitative models that omit the extensive margin. For a given negative labor demand shock ($\Delta\tilde{\zeta}^s_{t+1} < 0$), a standard model lacking the entry margin will mischaracterize the equilibrium adjustment: it will overstate wage responses and understate quantity responses.

This effect is state-dependent. The marginal contribution of entrants to the aggregate supply elasticity (Lemma 1) depends on the "slack" $E^s_t(1 - E^s_t)$. If a sector is already deeply depressed and entrant probabilities are near zero ($E^s_t \approx 0$), this margin's attenuating effect weakens. Back-of-the-envelope calculations, using our estimates and observed worker flows from the early 2000s, confirm that this was not the case; the entry margin remained active and empirically relevant, serving to amplify quantity pass-through and attenuate wage pass-through.[7]

Part (c) of the proposition points to a calibration wedge for quantitative studies of the decline of manufacturing. models that ignore the elastic entry margin will *systematically overestimate* the magnitude of underlying shocks. A common approach to studying the general equilibrium effects of the "China shock" (Autor et al., 2013) is to find the trade-cost reductions that replicate observed U.S. import penetration from China (usually through an instrument). Viewed through the "reduced exchange model" lens of Adao et al. (2017), any such import shock is observationally equivalent to a negative sectoral labor demand shock ($\Delta\tilde{\zeta}^s_{t+1} < 0$). Proposition 2-(c) implies that because our model features a more elastic labor supply, a smaller underlying shock is sufficient to generate the same import penetration and employment decline. A model that shuts down the entry margin must attribute the entire adjustment to less-elastic incumbents and therefore requires a larger calibrated shock to match the same facts. We quantify this wedge in Section 5.

4.3 Allocation of Adjustment Across Margins

Proposition 2 established that the elastic entry margin amplifies the aggregate quantity response. A natural next question is: how is the adjustment allocated between incumbents and entrants? The decomposition follows directly from Lemma 1. Each margin's share of the total employment response $\partial L^s_{t+1}/\partial\tilde{\zeta}^s_{t+1}$ is proportional to the product of its size, its elasticity ($1/\nu$ or $1/\nu^E$), and its choice slack $p(1-p)$. For small shocks, this allocation is approximately constant. But for discrete shocks, the allocation is nonlinear in the shock size:

Proposition 3 (Nonlinear allocation of adjustment)**.** *Consider a sequence of negative demand shocks to sector s.*

[7] Formally, the entry margin increases the aggregate supply elasticity x^s_t relative to an incumbents-only model whenever $\theta \leq \frac{\nu}{2\nu^E}\frac{E^s_t(1-E^s_t)}{q^s_t(1-q^s_t)}$, where $q^s_t := L_t^{-1}\sum_n L^n_t \mu^{ns}_t$ is the employment-weighted share of incumbents flowing into s. Using our estimates ($\nu \approx 2.44$, $\nu^E \approx 0.350$, $\theta = 0.97$) and observed entry-to-incumbent flow ratios $E^s_t/q^s_t \in (0.5, 2)$ over 1983–2019, this condition holds throughout the sample period. See Appendix A.2 for the derivation.

(a) Small shocks *(ΔV^s_{t+1} small): Adjustment is dominated by the margin with the highest slack-weighted elasticity. Given $\nu^E < \nu$, entrants drive a disproportionate share of the employment response whenever E^s_t is not near* 0 *or* 1.

(b) Large cumulative shocks *(V^s_{t+1} very low): Because entrants are more elastic, the entry share E^s_t approaches zero faster than incumbent inflow shares μ^{ns}_t, and entrant slack $E^s_t(1 - E^s_t)$ vanishes. The marginal adjustment to further shocks is therefore increasingly borne by the less elastic incumbent stock.*

Proposition 3 implies that the burden of adjustment shifts endogenously over the course of a protracted sectoral decline. Early on, forward-looking entrants respond strongly, absorbing most of the shock through reduced entry. But once entrants have largely abandoned the sector, the remaining adjustment falls on incumbents—who are less elastic and adjust primarily through retirement rather than switching. This is precisely the pattern documented in Section 2: young workers disproportionately avoid manufacturing entry (Fact 3), while incumbent manufacturing workers face earnings losses and low mobility, eventually exiting through retirement.

The proposition also implies that the combined impact of multiple shocks (e.g., trade and automation) is non-additive: the slack available on each margin depends on the total accumulated shock, not on each shock individually. Finally, it yields a testable cross-sectional prediction: local labor markets with a historically larger entrant share in manufacturing (higher baseline E^s_t) should exhibit more attenuated wage responses and larger employment responses to subsequent negative shocks.

4.4 Permanent shocks and the timing of adjustment

The previous results characterize one-period responses. We now ask how the entry margin shapes the *speed* of adjustment to a permanent shock—the setting most relevant for episodes like the China shock, automation, and the emergence of AI.

Consider an economy in steady state at $t = -1$. At $t = 0$, an unanticipated permanent negative demand shock $\Delta\xi^s < 0$ hits sector s for all $t \geq 0$. We linearize the supply and demand systems around the pre-shock steady state. Let $a \equiv \partial w^s / \partial L^s < 0$ be the slope of labor demand and $b \equiv \partial w^s / \partial \xi^s > 0$ the impact of the shifter, so that

$$\Delta w^s_t = a\, \Delta L^s_t + b\, \Delta \xi^s.$$

On the supply side, two forces govern the evolution of employment: the wage-responsive slope x from Lemma 1 (evaluated at steady-state values) and the persistence of the incumbent stock, $\rho = \theta \mu^{ss}$, where μ^{ss} is the steady-state stayer share. The linearized law of motion is

$\Delta L^s_{t+1} = \rho\, \Delta L^s_t + x\, \Delta w^s_{t+1}$. Substituting the demand equation and rearranging yields the transition equation

$$\Delta L^s_{t+1} = \underbrace{\frac{\rho}{1-ax}}_{\equiv \kappa}\, \Delta L^s_t + \underbrace{\frac{xb}{1-ax}}_{\equiv \gamma}\, \Delta \xi^s,$$

where $0 \leq \kappa < 1$ since $a < 0$ and $x > 0$. The persistence parameter κ governs the speed of transition: a smaller κ means faster convergence to the new steady state.

Proposition 4 (Front-loaded transition with entry)**.** *Let x denote the supply slope with an active entry margin ($E^s \in (0,1)$) and $x_{\text{noE}} < x$ the slope without it. Following a permanent shock $\Delta\xi^s < 0$, we have:*

(a) Faster Convergence (Front-Loading). The persistence parameter $\kappa = \rho/(1-ax)$ is strictly decreasing in x, so $\kappa < \kappa_{\text{noE}}$. The fraction of the long-run adjustment realized by horizon h, given by $1-\kappa^h$, is strictly larger for all $h \geq 1$.

(b) Muted Wage Pass-Through: The impact wage response is $\Delta w^s_1 = b\,\Delta\xi^s/(1-ax)$, and the long-run response is $\Delta w^s_\infty = b(1-\rho)\,\Delta\xi^s/(1-ax-\rho)$. Both are strictly decreasing in $|\cdot|$ with respect to x.

(c) Dynamic Decomposition and Front-loading: Entrants at $t = 0$ anticipate the full path of lower wages and immediately reduce entry into sector s, absorbing a large share of the employment adjustment on impact. As L^s_t falls and wages partially recover, the entry share E^s_t gradually rises from its trough toward a new, lower steady state E^s_∞.

Proof. (Sketch, see Appendix for full proof.) The results follow from the derivatives of κ and the wage-response term with respect to x: since $a < 0$ and $\rho > 0$, the persistence parameter ρ satisfies $\frac{\partial\rho}{\partial x} = \frac{a\rho}{(1-ax)^2} < 0$. Moreover, since $a < 0$ and $b > 0$, the derivative of the wage response magnitude is $\partial|\Delta w^s_1/\Delta\xi_s|/\partial x = -ab/(1-ax)^2 < 0$. From Lemma 1, an active entry margin ($x_{\text{ent}} > 0$) strictly increases x, proving the first two parts. The fraction of adjustment is $\Delta L^s_h/\Delta L^s_\infty = (1-\kappa^h)$, which is decreasing in κ. Result (3) follows from the decomposition of x in Lemma 1 and the evolution of $E_s(1-E_s)$ as E_s approaches a boundary. □

The economic intuition is straightforward. The entry margin provides a large, immediate source of supply elasticity. Forward-looking entrants respond to the full anticipated wage path, producing a sharp quantity adjustment at the onset of the shock and reducing the initial wage decline needed to clear the market. Without this margin, the entire burden falls on less-elastic incumbents, requiring a larger wage drop and a more protracted employment adjustment governed by ρ alone.

Finally, Proposition 4 connects to the nonlinear logic of Proposition 3: if the shock is sufficiently large, entry collapses and slack vanishes on the extensive margin. The remaining

adjustment is then gradually borne by incumbents, producing the slow tail of decline observed in the data.

Summary. The theory section delivers three results. First, baseline flows and the two estimated elasticities are sufficient statistics for counterfactual dynamics—cost levels are not needed (Proposition 1). Second, the elastic entry margin amplifies employment responses, attenuates wage responses, and implies that models without entry require larger calibrated shocks to match the same employment paths (Proposition 2). Third, the entry margin front-loads adjustment and endogenously shifts the burden from entrants to incumbents as shocks accumulate (Propositions 3–4). We now quantify these mechanisms through a calibrated model of the aggregate U.S. economy and a 50-state regional model in Section 5.

5 Quantitative Analysis: Adjustment to Shocks

Building on the theoretical insights, this section employs quantitative simulations to assess the importance of the entry margin in shaping labor market dynamics and the welfare consequences of trade shocks. We structure our analysis in two parts. First, we use a calibrated model of the aggregate US economy to provide a clean test of our theory, isolating the mechanical differences between our baseline model (with $\nu^E < \nu$ and cohort replacement) and the canonical model (which abstracts from entrants, $\theta = 1, N_t = 0$). Second, we embed the framework into a 50-region model to explore how these mechanisms interact with observed cross-state heterogeneity.

5.1 Quantitative Model Setup

We first outline the general equilibrium framework, which nests both the aggregate and multi-region models. We embed the labor supply block from Section 3 into a small open economy model where labor is mobile across sectors $s \in \{1, ..., S\}$ within a region $i \in \{1, 2, \cdots, I\}$, but immobile across regions.[8]

Labor Demand. We follow the demand-side structure of Artuç et al. (2010). As small open economies, all regions face exogenous world prices P^s_{it}, which we normalize to 1 in the initial steady state. Firms in each region i and sector s use a CES production function combining labor

[8]While inter-regional migration provides an additional adjustment margin in principle, we abstract from it, as regional migration flows are orders of magnitude smaller than sectoral reallocation flows. This choice is consistent with empirical evidence: Autor et al. (2021, 2025a) find that out-migration from regions exposed to the China shock was modest at best. Furthermore, Rodriguez-Clare et al. (2025) find that regional migration is highly inelastic and shutting it down in quantitative models leave counterfactual outcomes relatively unchanged. Olney and Thompson (2024) document that this decline of migration is a recent phenomenon driven by housing prices.

L_{it}^s and a fixed factor K_i^s (e.g., capital/land), normalized to $K_i^s = 1$, with productivity A_i^s:

$$y_{it}^s = A_i^s(\alpha_i^s(L_{it}^s)^{\rho^s} + (1-\alpha_i^s)(K_i^s)^{\rho^s})^{1/\rho^s}. \tag{14}$$

Firms are price-takers in goods and labor markets. Profit maximization implies that labor earns its marginal product, yielding the inverse labor demand function:

$$w_t^s = P_t^s \alpha^s A^s (L_t^s)^{\rho^s - 1}(\alpha^s (L_t^s)^{\rho^s} + (1-\alpha^s))^{\frac{1}{\rho^s}-1} \tag{15}$$

Labor Supply. Labor supply is governed by the entrant-incumbent model in Equations (2)-(5). The framework features incumbents with mobility parameter ν and survival probability θ, and entrants with mobility parameter ν^E and mass N_t. We use our estimated values for ν and ν^E from Section 3.2 (e.g., $\nu \approx 4.75, \nu^E \approx 0.71$). In our baseline, we assume the entrant mass N_t perfectly replaces the retiring worker mass, $N_t = (1-\theta)L_t$, keeping a constant aggregate labor force in each state.[9]

Equilibrium and Transition Dynamics. An equilibrium is a sequence of wages $\{w_{it}^s\}$, labor allocations $\{L_{it}^s\}$, and continuation values $\{V_{it}^s\}$ that satisfies firm optimization, worker optimization, and labor market clearing in each region i and sector s for all t.

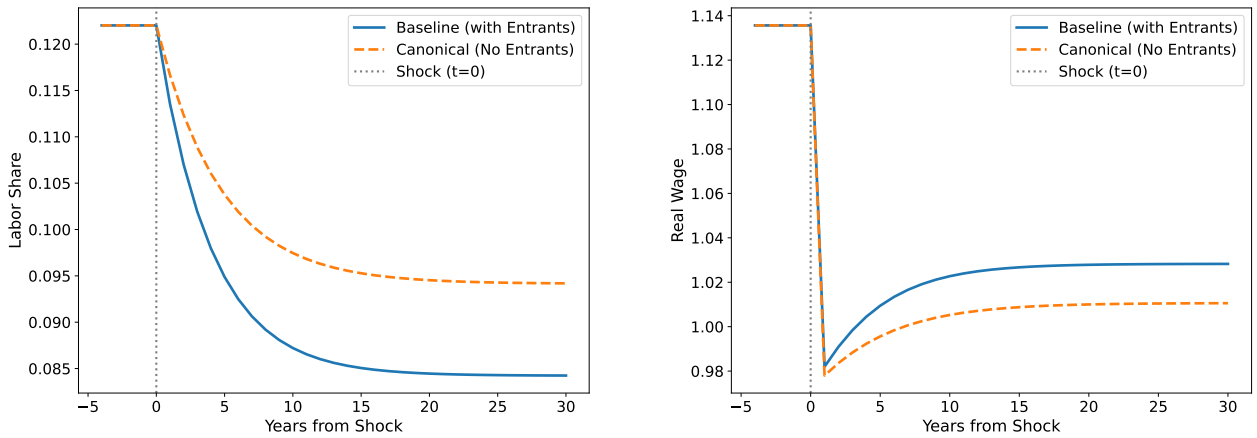
5.2 Testing the Mechanism: Stylized Shock in Aggregate Economy

To cleanly test the key predictions from Section 4, we first simulate an unexpected, permanent drop in the world price of manufacturing goods in a one-region model ($I = 1$) calibrated to the U.S. aggregate economy.

Calibration. We compare two models, both calibrating the labor demand-side parameters $(A_i^s, \alpha_i^s, \rho^s)$ to match aggregate sectoral wages, labor shares, and employment shares for the US in the year 2000. The first is the *baseline model* which uses our full labor supply block with the empirically estimated elasticities for incumbents ($\nu \approx 4.75$) and entrants ($\nu^E \approx 0.71$) and cohort replacement ($\theta < 1, N_t = (1-\theta)L_t$). The second model is the *canonical model* which abstracts from entry by setting $\theta = 1$ and $N_t = 0$. To isolate the adjustment dynamics and ensure a comparable starting point, we simulate both models from the identical initial labor allocation L_0, which is set to the observed 2000 employment shares. This experiment allows us to isolate the precise effect of the entry margin.

[9] We are working on extensions where (1) we match the realized entrant mass, and (2) entry into regions is endogenous – following empirical evidences by Autor et al. (2025b).

Figure 4: Employment and Wage Dynamics following shock



(a) Manufacturing employment (share)

(b) Manufacturing real wage

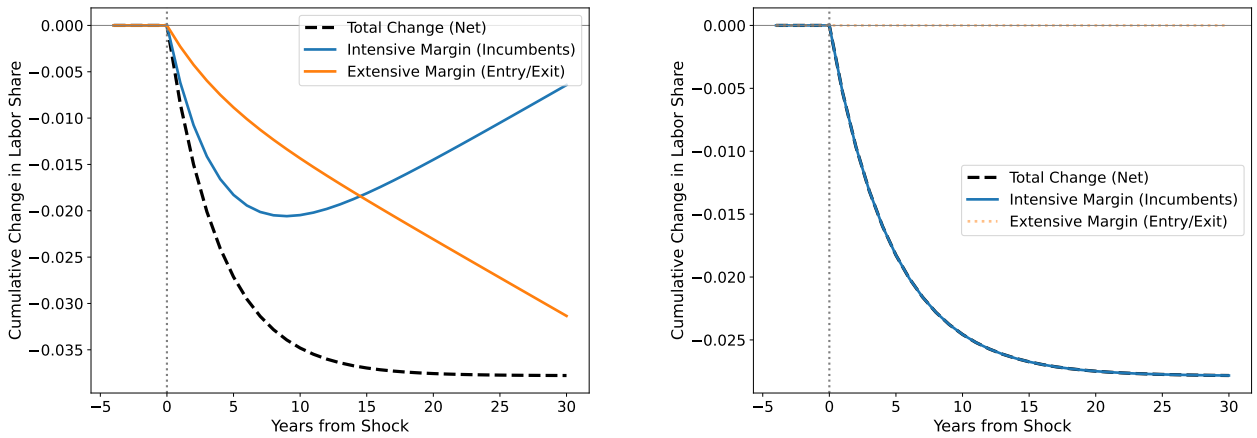
One shock. We solve for the perfect-foresight transition path for both models assuming a 20% drop in the world price of manufacturing goods (P^{mfg} falls from 1 to 0.8).

First, we examine the aggregate wage and employment dynamics, which provide a quantitative test of Proposition 2. Figure 4 plots the transition paths for both models. The results confirm the predicted trade-off: incorporating entrants feature both quantity amplification and wage attenuation. The left panel shows quantity amplification: the decline in manufacturing employment is not only faster and deeper in the baseline model, but the total long-run employment drop is larger for the *same* 20% price shock. The right panel shows the wage attenuation: the manufacturing wage falls less in the baseline model than in the canonical model, both on impact and in the new steady state. The intuition is precisely as the theory suggests: because the highly elastic entry margin provides a large quantity response, the market can clear with a smaller equilibrium price (wage) adjustment. Models omitting this margin will overstate wage declines and understate the quantity adjustment.

Second, we decompose the decline to identify who adjusts, mirroring the empirical facts from Section 2. Figure 5 illustrates the cumulative change attributable to the intensive margin (incumbent switching) versus the extensive margin (net entry/exit). The results provide validation of our motivating facts. In the baseline model, the extensive margin – a collapse in new entry combined with incumbent retirements – drives the vast majority of the adjustment. The intensive margin plays some role but diminishes, despite the fact that only a small fraction $(1-\theta)$ enter, and the bulk of the population are incumbents (θ). This is consistent with the low estimated elasticity of incumbents.

Nonlinearity in Shock Magnitude. To test the *nonlinear* predictions of Corollary **??**, we now

Figure 5: Decomposition of Manufacturing Decline (Cumulative)



(a) Baseline Model (with Entrants)

(b) Canonical Model (No Entrants)

simulate both models over a grid of 40 permanent manufacturing price shocks, ranging from a 1% to 40% decline. For each shock, we decompose the employment adjustment in manufacturing into the shares attributable to the extensive (entrant) and intensive (incumbent) margins at different horizons.

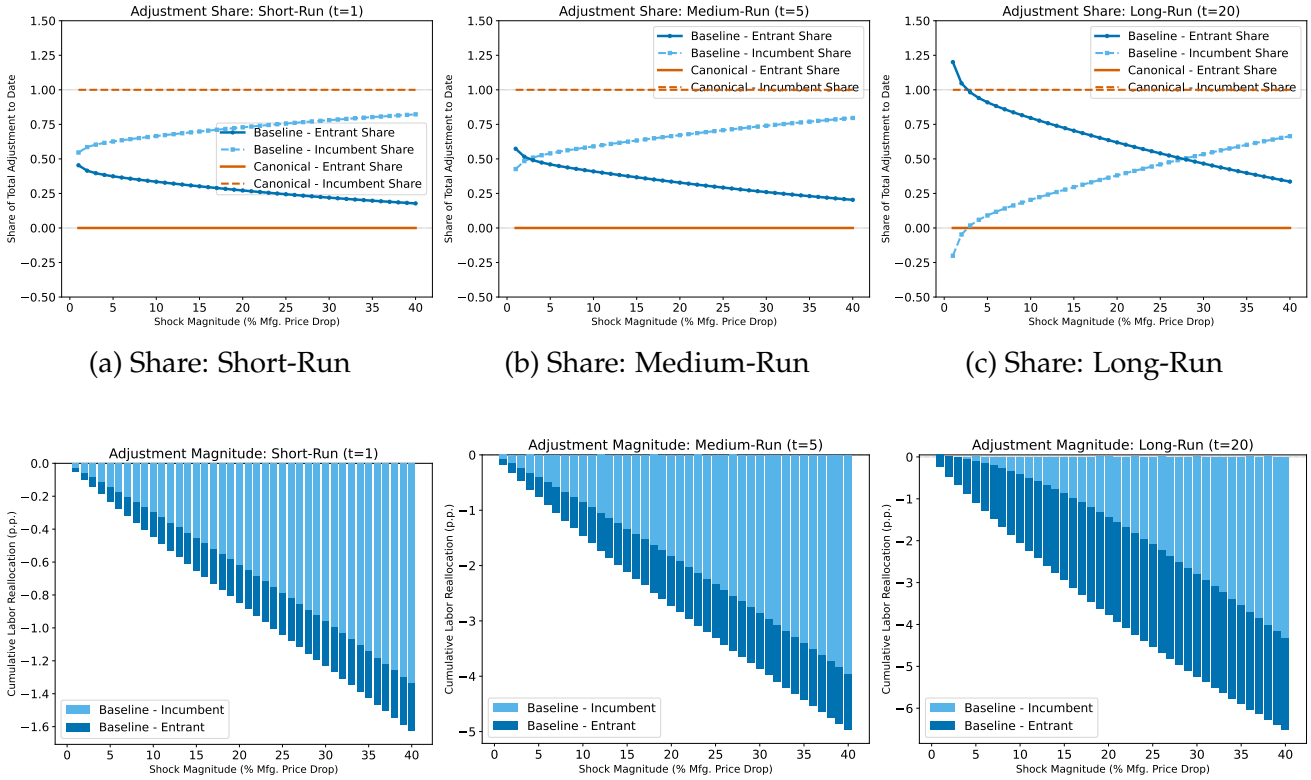
Figure 6 plots the plots the decomposition of adjustment shares (top row, panels a-c) and total magnitudes (bottom row, panels d-f). The canonical model provides a clear benchmark. In this case, with $\theta = 1$, the entrant margin is nonexistent, and 100% of the adjustment is by definition borne by incumbents. The baseline model, incorporating the entry margin, provides a test of our theory.

For small shocks, the margin with higher elasticity have a larger effect. In the short-run (panel a), the extensive (entrant) margin accounts for half of the adjustment. This result confirms the first part of Corollary 1. Specifically, when slack is high for both margins, the high-elasticity entrants drive a large share of the response – the only "limit" being the raw number of entrants (N_t).

As the shock magnitude increases (moving to the right in each panel), the "entrant slack" defined as $E_t^s(1 - E_t^s)$ diminishes. The manufacturing sector becomes so unattractive that the entrant probability E_t^s approaches zero. The plots illustrate this mechanism clearly. The entrant share of adjustment falls monotonically with the shock size. Consequently, the marginal adjustment is increasingly borne by the less elastic, but larger, stock of incumbents, whose share of the adjustment rises accordingly.

The figure also reveals a nonlinear, dynamic interplay between the margins, resulting in *overshooting* over time. An adjustment share greater than 100% (or below 0%) indicates that the two margins are working in opposite directions. This is evident in the long-run (panel c)

Figure 6: Decomposition of Manufacturing Adjustment by Shock, Horizon, and Margin



(a) Share: Short-Run (b) Share: Medium-Run (c) Share: Long-Run

(d) Magnitude: Short-Run (e) Magnitude: Medium-Run (f) Magnitude: Long-Run

Note: This figure plots the decomposition of cumulative employment adjustment in manufacturing following price shocks of varying magnitudes. The top row (a-c) plots the share of adjustment attributable to entrants (extensive) vs. incumbents (intensive). The bottom row (d-f) plots the total magnitude of adjustment in percentage points, decomposed into entrant and incumbent contributions. Short-run refers to margin in 1 year; medium-run refers to 5 years; long-run refers to 20 years after the shock.

for small shocks, where the entrant share exceeds 100% while the incumbent share becomes negative. This dynamic demonstrates that high-elasticity entrants flee the sector rapidly that they stabilize the wage, which in turn temporarily draws a small number of low-elasticity incumbents into manufacturing. The net job loss is the sum of these opposing flows. This overshooting effect suggests a reason why incumbent reallocation can contribute negligibly to the net adjustment empirically (Section 2). With the entrants responding quickly, the intensive margin may not respond, or may even move marginally in the opposite direction, offsetting the extensive margin adjustment.

This experiment illustrates that incorporating entrant-incumbent heterogeneity changes the predicted adjustment path both qualitatively and quantitatively. This framework aligns well with stylized facts and highlights potential biases in canonical models regarding wage responses and welfare incidence.

5.3 Cross-State Evidence: Heterogeneity in Adjustment

Having isolated the aggregate mechanism, we now explore how regional heterogeneity shapes the response to trade shocks. We use a multi-region version of the model calibrated to the 50 U.S. states.

Model Setup and Calibration. We embed the framework from Section 5.1 into a 50-region model ($I = 50$).[10] The labor demand-side parameters (A^s, α^s, ρ^s) and supply-side costs (χ^{sn}, C^n) are calibrated to match regional sectoral wages, labor share of revenue, sector share of GDP, and sectoral employment shares for each state in the year 2000. We again compare the Baseline (entrant) model against the Canonical (no-entrant) model, which has the same calibrated parameters. We solve for the perfect-foresight transition path in all 50 states following the same 20% drop in manufacturing prices.

Findings. Figure 7 plots the cross-state results for the Baseline model (left column), highlighting the role of regional heterogeneity. The right column presents results from the Canonical model for comparison.

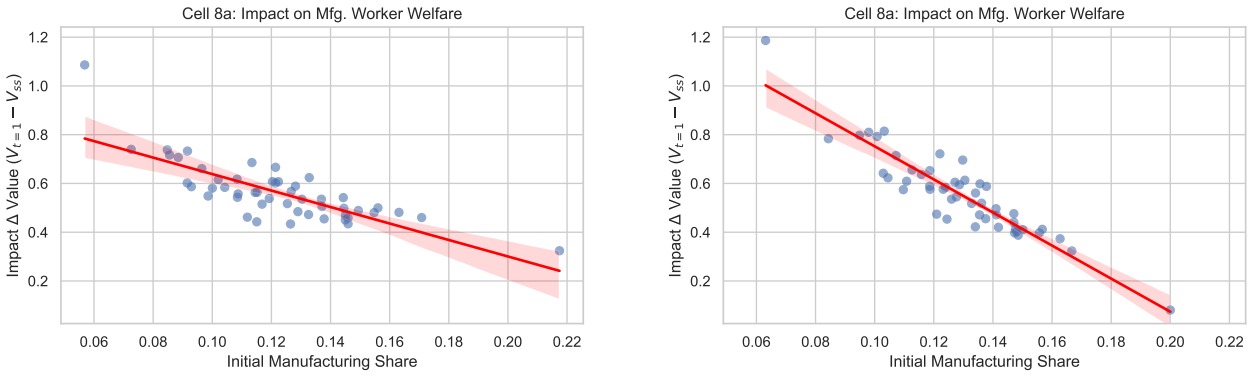
The entrant margin fundamentally alters the relationship between initial manufacturing dependence and the shock's cost. In the Baseline model, the negative correlation between a state's initial manufacturing share and the welfare loss on impact is significantly muted (Panel 7a, left). The high-elasticity entrant margin acts as a "buffer" that absorbs the initial shock, thereby protecting incumbent welfare. This welfare protection is achieved via a stronger quantity response. Panel 7b shows that high-manufacturing states experience a much larger total reallocation (a steeper negative slope), which aligns with Proposition 2.

The adjustment dynamics are also very different. Under the Baseline model, adjustment speed is relatively uniform across states (Panel 7c). This uniformity stems from the high-elasticity entrant and retirement margin. The composition of adjustment (Panel 7d) reflect a striking pattern. In states with a larger initial manufacturing share, the model predicts a positive net flow from the intensive margin. This occurs because the extensive margin (entrants and retirees) responds so strongly that it over-compensates for the shock. This allows existing workers with lower elasticities to fill the few remaining manufacturing jobs, even as the net employment adjustment remains negative.

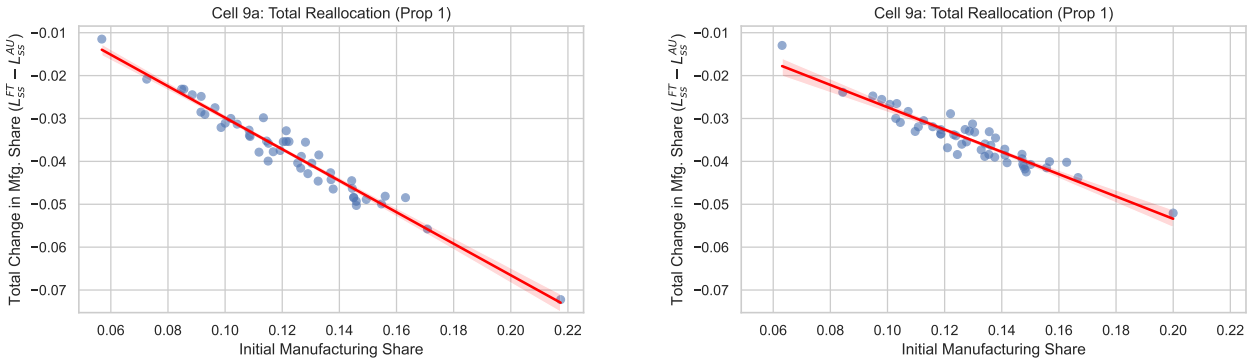
We next examine the characteristics of the service sector to which new workers adjust. Figure 8 plots the total manufacturing employment decline against the initial service sector wage in our simulation. We find a negative relationship between decline and service jobs. The largest manufacturing declines occur precisely in states where the initial service sector wage (a proxy

[10] We are in progress with incorporating a more full-fledged workhorse trade model with input-output linkages, imperfect substitution, and leaving the small open economy.

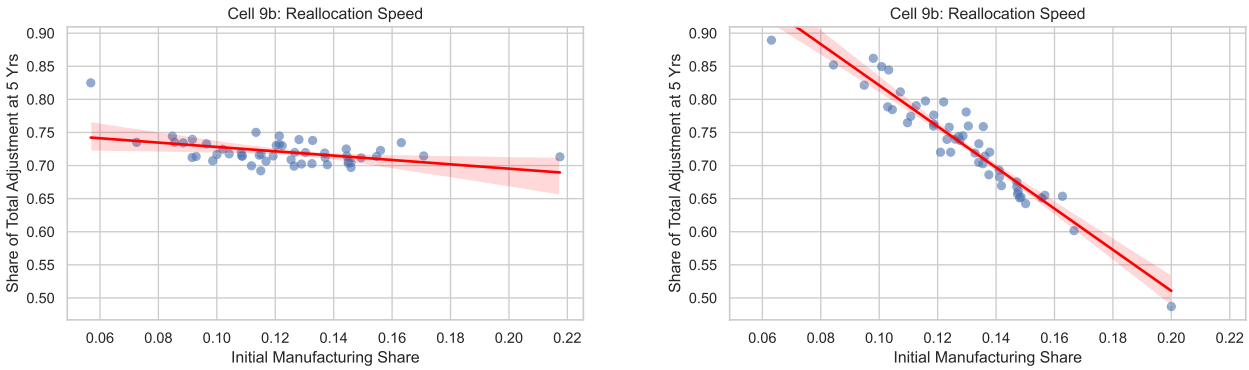
Figure 7: Cross-State Heterogeneity: Baseline vs. Canonical Model



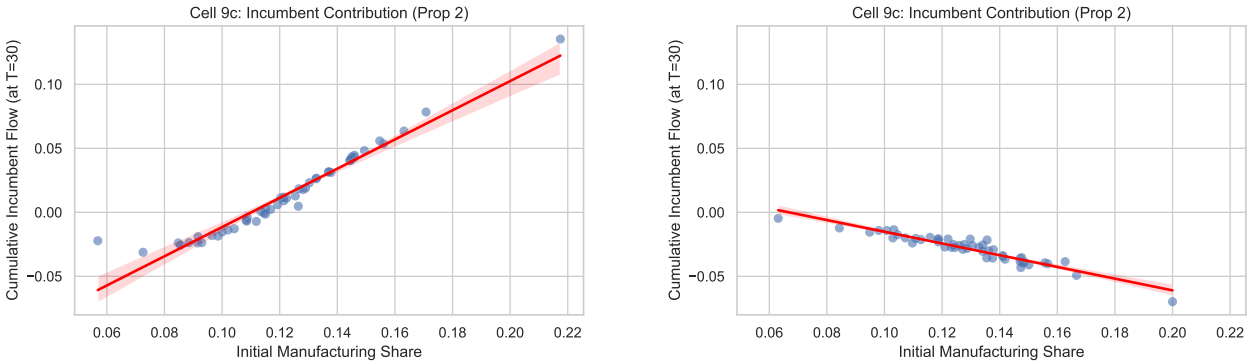
(a) Impact on Mfg. Worker Welfare



(b) Total reallocation (Prop 1)

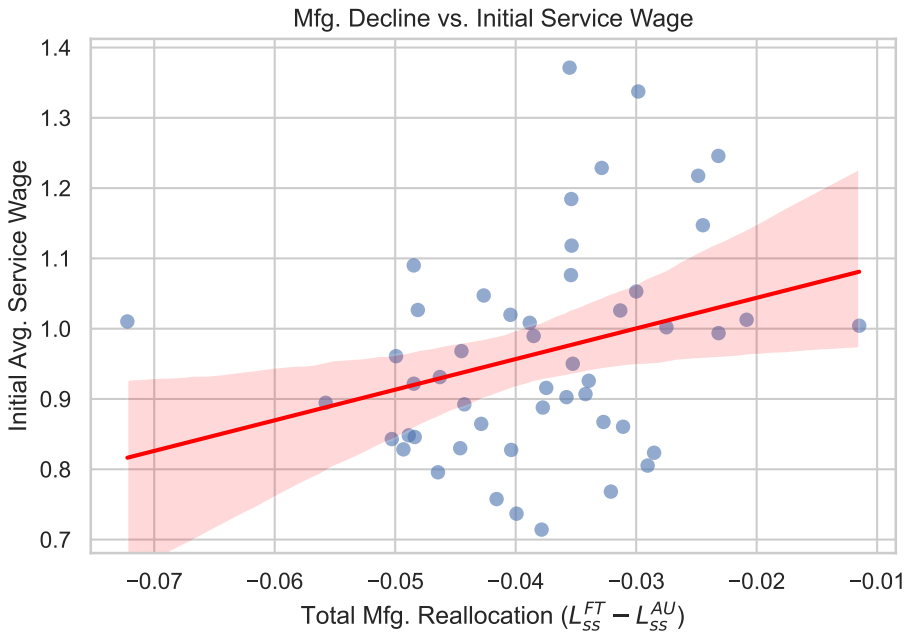


(c) Reallocation Speed (Share of Adj. at 5 Yrs)



(d) Intensive Margin Contribution in MFG decline (Prop 2)

Figure 8: Manufacturing Decline vs. Initial Service Sector Wage



for A^n) is lowest. This implies that the "rebound" in service jobs, which occurs mechanically as manufacturing declines, is concentrated in regions where existing service sector jobs are of lower quality. This result aligns with the empirical findings of Autor et al. (2025b), which find that regional recovery happened through increases in low-premium service jobs.

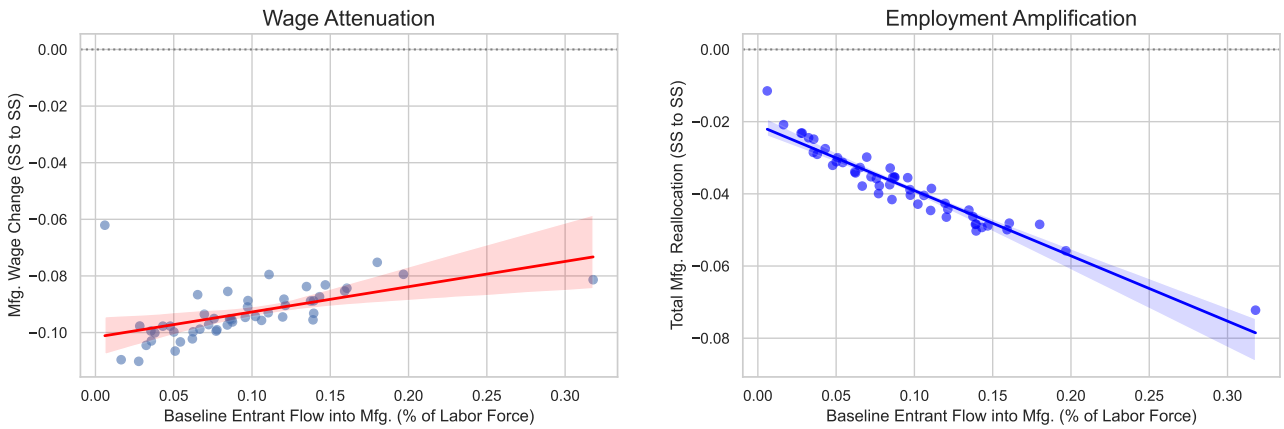
Finally, we test the prediction in Section 4: local labor markets with a historically larger share of entrants in manufacturing (i.e., a higher baseline E_t^s) should exhibit more attenuated manufacturing wage responses and, simultaneously, larger employment responses to a negative shock. Figure 9 confirms this prediction. We plot the simulated wage and employment responses against the initial baseline entrant share for all 50 states. The results show that, in response to the same aggregate price shock, regions with higher initial entry rates experience smaller wage declines and larger employment adjustments.

5.4 Calibration of the China shock

We now discipline the time path of the manufacturing price index so that the model replicates the rise in manufacturing import penetration associated with the China shock. Our strategy mirrors the inversion in Caliendo et al. (2019): rather than estimating high-dimensional fundamentals, we use the equilibrium conditions to find the sequence of exogenous shifters that matches observed trade outcomes.

Target and Procedure. Our target is the change in import penetration of Chinese manufactur-

Figure 9: Test of Mechanism: Baseline Entrant Share vs. Shock Response



(a) Wage Attenuation

(b) Employment Amplification

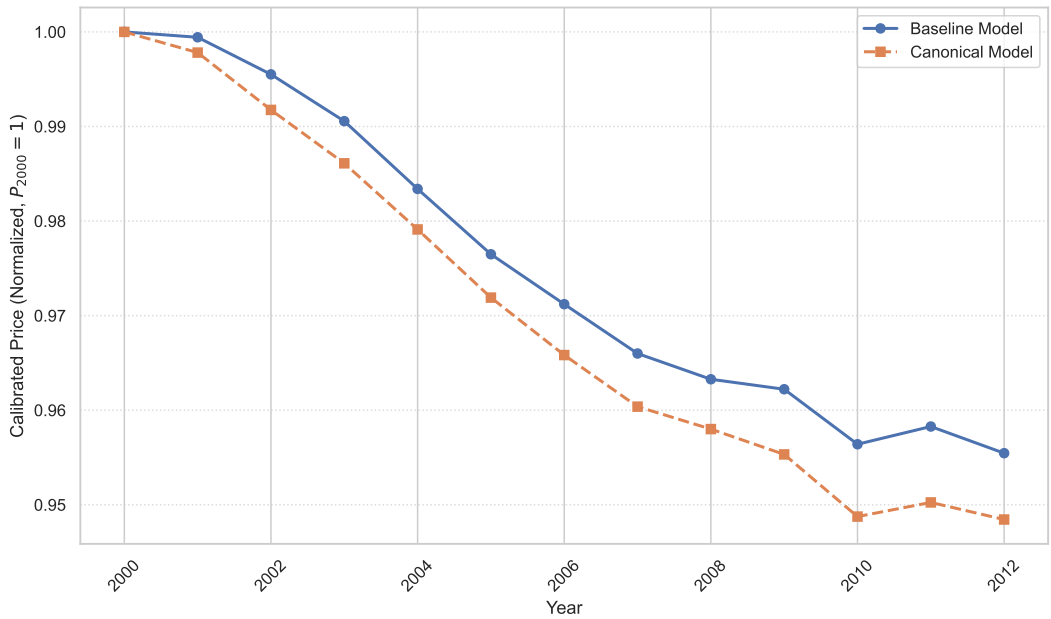
ing goods in the US (ΔIP_t) over the 2000-2012 window.[11] In our small open economy framework, the exogenous shifter is the sectoral goods price, P_t^s. We therefore invert a standard import-demand system to find the time series of equivalent price shifters $\{\xi_t^{Mfg}\}_{t=2000}^{2012}$ that would generate the observed ΔIP_t path. We then implement this sequence as the exogenous path for P_t^{Mfg}. We perform this inversion twice: first using our baseline model (with $\nu^E < \nu$) and second using the canonical model ($\theta = 1, N_t = 0$). This parallel calibration isolates the calibration wedge highlighted by Proposition 2: when the extensive margin is active and more elastic, the implied price changes needed to rationalize the data must be smaller.

Findings. As predicted, the model with the more elastic aggregate labor supply (the Baseline) should require a smaller price shock to rationalize the same data. Figure 10 plots the recovered price sequences. The results confirm our prediction: the price path inferred by the Baseline model is uniformly higher (less negative) than the path required by the Canonical model. By 2012, the price drop (the "China shock") implied by the Baseline model is 15.7 % smaller than the drop implied by the Canonical model. This finding is a direct consequence of the elastic entrant margin. Because the Baseline model's aggregate supply curve is more elastic, a milder price decline is sufficient to generate the same observed rise in import penetration.

Thus, explicitly incorporating worker entry and exit will lead to different welfare estimates for two primary reasons. First, conditional on the same shock ($\xi_t^s < 0$), our model predicts that wages move less and quantities adjust more, leading to smaller welfare losses. Second, conditional on matching the same data (ΔIP), the calibrated shock itself is smaller. Because

[11] Since we assume a small open economy, the model does not distinguish imports across origins. We calibrate the model so that the change in the aggregate manufacturing import penetration, $(C_t - Y_t)/C_t - (C_{2000} - Y_{2000})/C_{2000}$, matches the observed increase in import penetration specifically from China. The data target is $IM_t^{\text{mfg,china}}/(Y_t^{\text{mfg}} + IM_t^{\text{mfg}} - EX_t^{\text{mfg}}) - IM_{2000}^{\text{mfg,china}}/(Y_{2000}^{\text{mfg}} + IM_{2000}^{\text{mfg}} - EX_{2000}^{\text{mfg}})$, for $t =$ 2001-2012.

Figure 10: Calibrated Manufacturing Price Paths (P_t^{Mfg})



both biases – the adjustment channel and the calibration wedge – move in the same direction, a model featuring the extensive margin will estimate smaller welfare costs (and larger benefits) from trade shocks compared to a canonical model that omits it.

6 Welfare and Policy

The entrant–incumbent distinction has direct implications for policy. Incumbents face high switching costs and respond sluggishly to wage differentials; entrants face low entry costs and are highly elastic in their initial sector choice. This asymmetry means that the two margins are not interchangeable, and that the ranking of instruments depends sharply on what the planner is trying to achieve.

We organize the analysis around three questions. First, is there a welfare rationale for intervention at all? We show that absent distortions, the answer is no: the decentralized equilibrium is constrained efficient, and the frictional slow adjustment it implies is optimal given workers' preferences (Section 6.1). Second, if the goal is one of slowing the decline of a particular sector—motivated by political economy, national security, or agglomeration forces outside the scope of our model—which margin should policy target? We show that entry-targeted instruments have systematically higher leverage than incumbent-targeted ones, and characterize when this ranking is large (Section 6.2). Third, once we introduce genuine distortions—downward nominal wage rigidity and local recession externalities—there is a real efficiency case for intervention, and the same entry-elasticity asymmetry that was irrelevant under laissez-faire now deter-

mines the optimal instrument mix (Section 6.3).

[This section is under construction.]

6.1 Constrained efficiency: a benchmark

The logit choice structure has a useful property: it admits an exact *entropy* planner representation (Anderson et al., 1988; Hofbauer and Sandholm, 2002).

The key identity is that for any vector $a \in \mathbb{R}^S$ and $\nu > 0$,

$$\nu \log\Big(\sum_n e^{a_n/\nu}\Big) \;=\; \max_{p\in\Delta}\Big\{\sum_n p_n a_n + \nu H(p)\Big\}, \qquad H(p) \equiv -\sum_n p_n \log p_n, \tag{16}$$

with unique maximizer $p_n = e^{a_n/\nu} / \sum_k e^{a_k/\nu}$ — exactly the logit rule. In words: the expected utility a logit agent achieves equals the maximum of expected payoffs plus an entropy bonus that captures the value of preference heterogeneity. A constrained planner who cannot observe idiosyncratic taste shocks but chooses allocation probabilities $\{\mu_t^{sn}, E_t^n\}$ subject to the law of motion

$$L_{t+1}^n = \theta \sum_s \mu_t^{sn} L_t^s + N_t E_t^n \tag{17}$$

faces exactly the right-hand side of (16) as its objective at each margin.

Applied to our entrant–incumbent block where the two margins have different dispersion parameters ν (incumbents) and ν^E (entrants), the incumbent and entrant value functions satisfy

$$V_t^s \;=\; u(w_t^s) + \max_{\mu_t^{s\cdot}\in\Delta}\Big\{\sum_n \mu_t^{sn}\big(\beta\theta V_{t+1}^n - \chi^{sn}\big) + \nu\, H(\mu_t^{s\cdot})\Big\}, \tag{18}$$

$$\mathbb{E}\big[\mathcal{V}_t^{ent}\big] \;=\; \max_{E_t\in\Delta}\Big\{\sum_n E_t^n\big(\beta V_{t+1}^n - C^n\big) + \nu^E H(E_t)\Big\}, \tag{19}$$

and in each case the unique maximizer coincides with the decentralized logit rule.[12] Note that the entrant entropy term $\nu^E H(E_t)$ is not redundant: because $\nu^E \neq \nu$ in general, entrants and incumbents contribute differently to planner welfare, and this distinction will matter for the policy analysis that follows.

Proposition 1 (Constrained efficiency)**.** *Absent externalities, missing markets, or wage rigidities, the decentralized allocation (μ_t, E_t) maximizes utilitarian welfare subject to (17). Taste shocks and mobility/entry costs enter as worker preferences rather than correctable wedges, so the planner cannot improve on the market outcome.*

This benchmark, which is essentially a consequence of (18)–(19), is the precise sense in which the baseline model is a *frictional adjustment* model but not an *inefficiency* model. The

[12] The identity (16) follows immediately from the first-order conditions of the right-hand side: the Lagrangian for the simplex constraint gives $p_n \propto e^{a_n/\nu}$. See Donald et al. (2025b) for a generalized derivation in spatial settings.

remainder of this section studies two departures from it: a planner with a sector-preservation objective (Section 6.2) and an explicit local recession distortion that breaks the efficiency result (Section 6.3).

6.2 A sector-preservation planner

Policy goal: seek to maximize L^s_{t+1} subject to welfare cost of interventions (on C^s, χ^{ns}). Much more efficient to target entry.

6.3 Second-best Policy: local recessions

Wedge (wage rigidity, local recession, $\cdots$) that makes social value of work higher than private value. Incentive to intervene.

Microfoundation: Linear-quadratic problem with nominal rigidity.

6.4 Policy implications?

The distinction between low-elasticity incumbents and high-elasticity entrants yields different predictions for policy effectiveness.

- Incumbent-Targeted Policies (e.g., Trade Adjustment Assistance (TAA), retraining subsidies). These policies aim to lower incumbents' switching costs (χ^{sn}). Our model predicts modest effects because they target the low-elasticity group (ν is large). While potentially improving welfare for treated individuals, their impact on aggregate reallocation speed might be limited, consistent with mixed empirical evidence on TAA effectiveness.

- Entrant-Targeted Policies (e.g., training for new skills, incentives for entering growth sectors): These policies could lower entry costs (C^n) into non-manufacturing sectors or raise the relative value (V^n_{t+1}) of those sectors. Given entrants' high elasticity ($1/\nu^E$), such policies could significantly accelerate structural transformation by steering new cohorts away from declining sectors and towards expanding ones.

- Place-Based Policies (e.g., investment credits in distressed regions): These policies aim to boost local labor demand ($\tilde{\zeta}^s_{it}$). Our model suggests they might attract high-elasticity entrants (including young workers and potentially immigrants) relatively quickly, helping stabilize local employment, as seen empirically in Autor et al. (2025b). However, their benefit to low-elasticity incumbents might be smaller, potentially creating distributional tensions within the targeted regions.

- Trade Policy (e.g., Tariffs): Temporary tariffs are predicted to be particularly ineffective. Forward-looking, high-elasticity entrants heavily discount future reversals, so temporary protection offers little incentive for them to enter the protected sector. Incumbents respond weakly due to low elasticity. Temporary tariffs might mainly delay necessary reallocation without substantially boosting long-run employment in the protected sector.

In essence, the model suggests that policies aiming to accelerate adjustment might be more effective when focused on guiding the choices of the more responsive extensive margin (entrants), while policies aimed at mitigating hardship might need to focus on supporting the less mobile intensive margin (incumbents), potentially through transfers rather than solely retraining, acknowledging their lower propensity to switch sectors. The findings resonate with the "places vs. people" distinction in Autor et al. (2025b), suggesting that successful place-based recovery may rely heavily on attracting new people (entrants) rather than primarily retooling existing residents (incumbents).

7 Concluding remarks

We show the extensive margin driven by generational turnover – fewer young entrants replacing retirees – is the primary driver of the long-run decline in U.S. manufacturing, overshadowing the intensive margin of incumbent workers switching sectors. We formalize this mechanism in a quantitative general equilibrium model featuring heterogeneous mobility elasticities, where entrants are calibrated to be significantly more responsive to wage signals than incumbents.

Incorporating this entrant-incumbent distinction yields quantitative predictions that differ starkly from canonical models. First, the high-elasticity entrant margin provides a "buffer" against shocks, leading to wage attenuation (smaller wage declines) and quantity amplification (larger employment adjustments). This mechanism mutes the welfare losses for incumbents, particularly in regions with high initial manufacturing dependence. Second, the model replicates nonlinear adjustment paths: the entrant share of adjustment is largest for small shocks but diminishes as large shocks make the sector unattractive, forcing the less-elastic incumbent margin to bear a larger share of the adjustment. Third, when calibrating the model to match observed trade flows, a model with entry-exit requires smaller "China shock" price decline than a homogeneous-agent model. This suggests canonical models may systematically overestimate the magnitude of underlying shocks by omitting the elastic extensive margin.

These findings have direct policy implications. They suggest that policies targeting entrants, such as training incentives for new skills, may be more effective for accelerating structural change. Conversely, policies aiming to support incumbents might be better focused on mitigating hardship given their low mobility, rather than primarily on inducing reallocation.

Our framework provides a tractable method for incorporating these generational dynamics into standard quantitative models of trade and structural change.

Several avenues for future research remain. Our analysis makes simplifying assumptions, such as abstracting from endogenous regional entry choice and other life-cycle dynamics like human capital accumulation. Furthermore, two open questions warrant a deeper investigation. First, the treatment of non-retirement movements into nonemployment requires refinement to distinguish between voluntary reallocation and involuntary exits, which have different welfare implications. Second, explicitly modeling unemployment and broader labor market frictions could provide a richer understanding of how these factors shape both incumbent reallocation and entrant decisions.

Addressing these issues is essential for a fuller picture of the incidence of trade shocks in labor market and structural transformation at large. The framework developed here, emphasizing generational entry and exit, offers a valuable lens for analyzing not only trade shocks but also other significant drivers of economic change, such as automation and shifts in sectoral demand. Understanding the distinct behavior of different cohorts within these transitions remains a critical task.

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Appendix

A Derivations

A.1 Derivation of Equations in Section 3

We provide the step-by-step derivation for the incumbent and entrant moment conditions (Equations 6 and 7) presented in Section 3.2.

Incumbent Estimating Equation

The derivation for the incumbent's Euler equation (Equation (6)) is as follows. We begin with the incumbent's logit probability of moving from sector s to sector n, as given by Equation (2):

$$\mu_t^{sn} = \frac{\exp\left(\frac{1}{\nu}\left(\beta\theta V_{t+1}^n - \chi^{sn}\right)\right)}{\sum_k \exp\left(\frac{1}{\nu}\left(\beta\theta V_{t+1}^k - \chi^{sk}\right)\right)}$$

The probability of staying in sector s (i.e., moving from s to s) is:

$$\mu_t^{ss} = \frac{\exp\left(\frac{1}{\nu}\left(\beta\theta V_{t+1}^s - \chi^{ss}\right)\right)}{\sum_k \exp\left(\frac{1}{\nu}\left(\beta\theta V_{t+1}^k - \chi^{sk}\right)\right)}$$

We take the log-ratio of these two probabilities, and the denominator cancels. Using the normalization that the cost of staying is zero ($\chi^{ss} = 0$), we arrive at the incumbent's log-odds ratio:

$$\log(\mu_t^{sn}/\mu_t^{ss}) = \frac{1}{\nu}\left[\beta\theta(V_{t+1}^n - V_{t+1}^s) - \chi^{sn}\right] \tag{A.1}$$

Next, we derive an alternative expression for the incumbent's ex-ante value function (Equation 3):

$$V_t^s = u(w_t^s) + \nu \log\left(\sum_k \exp\left(\frac{1}{\nu}\left(\beta\theta V_{t+1}^k - \chi^{sk}\right)\right)\right)$$

We can rearrange the expression for the stayer probability μ_t^{ss} to solve for the log-sum term:

$$\sum_k \exp\left(\frac{1}{\nu}\left(\beta\theta V_{t+1}^k - \chi^{sk}\right)\right) = \frac{\exp\left(\frac{1}{\nu}\beta\theta V_{t+1}^s\right)}{\mu_t^{ss}}$$

Substituting this back into the value function equation, we get a recursive expression for the value function:

$$V_t^s = u(w_t^s) + \beta\theta V_{t+1}^s - \nu \log(\mu_t^{ss}) \tag{A.2}$$

This identity, derived by Artuç et al. (2010), shows that the ex-ante value function can be written in terms of the current wage, the discounted future value, and an "option value" term $\nu \log(\mu_t^{ss})$.

We first derive an identity for the difference in continuation values by substituting the log-odds ratio (Equation A.1) into the value function difference derived from Equation (A.2):

$$\begin{aligned} V_t^n - V_t^s &= [u(w_t^n) - u(w_t^s)] + \beta\theta(V_{t+1}^n - V_{t+1}^s) - \nu(\log \mu_t^{nn} - \log \mu_t^{ss}) \\ &= [u(w_t^n) - u(w_t^s)] + [\nu \log(\mu_t^{sn}/\mu_t^{ss}) + \chi^{sn}] - \nu(\log \mu_t^{nn} - \log \mu_t^{ss}) \\ &= u(w_t^n) - u(w_t^s) + \chi^{sn} + \nu \log(\mu_t^{sn}/\mu_t^{nn}) \end{aligned}$$

Advancing this identity one period ($t \to t+1$) yields an expression for the future value difference:

$$V_{t+1}^n - V_{t+1}^s = u(w_{t+1}^n) - u(w_{t+1}^s) + \chi^{sn} + \nu \log(\mu_{t+1}^{sn}/\mu_{t+1}^{nn}) \tag{A.3}$$

We substitute this back into the log-odds ratio (Equation A.1):

$$\log(\mu_t^{sn}/\mu_t^{ss}) = \frac{1}{\nu}\left[\beta\theta\left(u(w_{t+1}^n) - u(w_{t+1}^s) + \chi^{sn} + \nu \log(\mu_{t+1}^{sn}/\mu_{t+1}^{nn})\right) - \chi^{sn}\right]$$

Simplifying and grouping the χ^{sn} terms gives the estimating equation:

$$\log(\mu_t^{sn}/\mu_t^{ss}) = \frac{\beta\theta}{\nu}[u(w_{t+1}^n) - u(w_{t+1}^s)] + \beta\theta \log(\mu_{t+1}^{sn}/\mu_{t+1}^{nn}) - \frac{1-\beta\theta}{\nu}\chi^{sn}$$

Adding the expectation error Ω_{t+1}^{sn} yields Equation (6).

Entrant Estimating Equation

The share of entrants choosing sector n is:

$$E_t^n = \frac{\exp\left(\frac{1}{\nu^E}\left(\beta V_{t+1}^n - C^n\right)\right)}{\sum_k \exp\left(\frac{1}{\nu^E}\left(\beta V_{t+1}^k - C^k\right)\right)}$$

Taking logs, we can express this as:

$$\log E_t^n = \frac{1}{\nu^E}\left(\beta V_{t+1}^n - C^n\right) - \log\left(\sum_k \exp\left(\frac{1}{\nu^E}\left(\beta V_{t+1}^k - C^k\right)\right)\right)$$

The log-sum term is common to all choices n and depends only on time t. We can therefore write the equation as:

$$\log E_t^n = \frac{\beta}{\nu^E} V_{t+1}^n - \frac{1}{\nu^E} C^n + \alpha_t' \tag{A.4}$$

where $\alpha'_t = -\log\left(\sum_k \exp(...)\right)$ will be absorbed by a time fixed effect in the estimation.

Substitute the incumbent's value function $V^n_{t+1} = u(w^n_{t+1}) + \beta\theta V^n_{t+2} - \nu \log(\mu^{nn}_{t+1})$ (Equation A.2 advanced one period) into (A.4):

$$\log E^n_t = \frac{\beta}{\nu^E}\left[u(w^n_{t+1}) + \beta\theta V^n_{t+2} - \nu \log(\mu^{nn}_{t+1})\right] - \frac{1}{\nu^E}C^n + \alpha'_t$$

This gives the expression:

$$\log E^n_t = \frac{\beta}{\nu^E}u(w^n_{t+1}) - \frac{\beta\nu}{\nu^E}\log(\mu^{nn}_{t+1}) + \frac{\beta^2\theta}{\nu^E}V^n_{t+2} - \frac{1}{\nu^E}C^n + \alpha'_t$$

To replace the V^n_{t+2} term, we advance Equation (A.4) one period and solve for $\frac{\beta}{\nu^E}V^n_{t+2}$:

$$\frac{\beta}{\nu^E}V^n_{t+2} = \log E^n_{t+1} + \frac{1}{\nu^E}C^n - \alpha'_{t+1}$$

We then rewrite the V^n_{t+2} term in the $\log E^n_t$ equation as $\beta\theta\left(\frac{\beta}{\nu^E}V^n_{t+2}\right)$ and substitute this expression in:

$$\log E^n_t = \frac{\beta}{\nu^E}u(w^n_{t+1}) - \frac{\beta\nu}{\nu^E}\log(\mu^{nn}_{t+1}) + \beta\theta\left[\log E^n_{t+1} + \frac{1}{\nu^E}C^n - \alpha'_{t+1}\right] - \frac{1}{\nu^E}C^n + \alpha'_t$$

Grouping the C^n terms and redefining the time fixed effect as $\alpha_t = \alpha'_t - \beta\theta\alpha'_{t+1}$, we get the final equation:

$$\log E^n_t = \frac{\beta}{\nu^E}u(w^n_{t+1}) + \beta\theta\log(E^n_{t+1}) - \frac{\beta\nu}{\nu^E}\log\mu^{nn}_{t+1} + \alpha_t - \frac{1-\beta\theta}{\nu^E}C^n \tag{A.5}$$

Adding the expectation error $\Omega^{n,E}_{t+1}$ gives the full equation.

Justification for Parameter Restriction. A direct estimation of Equation (A.5) faces three distinct challenges, leading us to adopt the behaviorally grounded restriction ($\nu^{\text{1-period}} = \nu^E$) as in the main text.

- **Structural Heterogeneity:** The parameter ν in the $\frac{\beta\nu}{\nu^E}$ term represents the entrant's expectation of their incumbent elasticity in the next period. The estimated ν (Table **??**) captures the average elasticity across all incumbents, including those with significant sector-specific tenure. It is overly restrictive to assume a first-period incumbent's elasticity mirrors this long-run average. A more plausible assumption is that early-tenure elasticity remains highly comparable to entrant elasticity ($\nu^{\text{1-period}} \approx \nu^E$). Imposing this restriction simplifies the $\frac{\nu}{\nu^E}$ ratio to 1.

- **Behavioral Plausibility:** Even if early-tenure elasticity is strictly higher than entrant elasticity, Equation (A.5) requires entrants to perfectly foresee and internalize this shift. This introduces a behaviorally questionable artifact: if $\nu \gg \nu^E$, agents would be artificially drawn to sectors with low retention rates (low μ^{nn}) to capitalize on the magnified variance of their future taste shocks. A more realistic mechanism aligns with projection bias (Loewenstein et al., 2003), where entrants project their current high-elasticity state onto their near-term future selves, perceiving their future elasticity as ν^E.

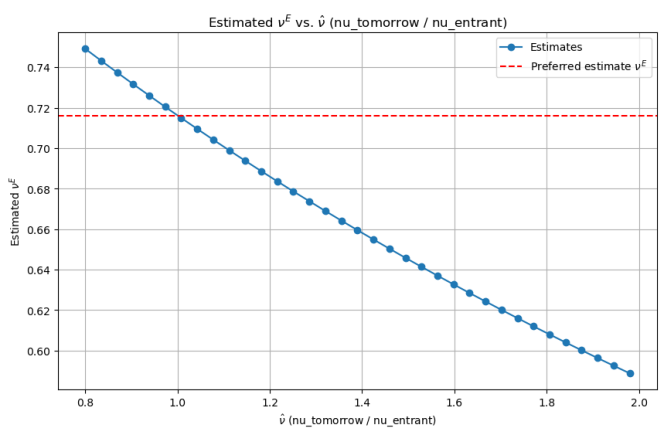
- **Empirical Identification:** Sectors with favorable unobservables typically exhibit both high wages (w_t^n) and high retention rates (μ_t^{nn}). Because these variables operate with opposite signs in the choice probability, estimating ν^E as a coefficient that responds simultaneously to both terms introduces collinearity and instability. Restricting the relative elasticity in option value stabilizes the moment conditions.

For these three reasons, we adopt the more stable and plausible specification by assuming $\frac{\nu^{\text{1-period}}}{\nu^E} = 1$ within the coefficient on the μ_{t+1}^{nn} term, yielding our final estimating equation:

$$\log E_t^n = \frac{\beta}{\nu^E} u(w_{t+1}^n) + \beta\theta \log(E_{t+1}^n) - \beta\theta \log \mu_{t+1}^{nn} + \alpha_t - \frac{1-\beta\theta}{\nu^E} C^n + \Omega_{t+1}^{n,E} \tag{A.6}$$

which give our preferred estimates of $\nu^E = 0.350$. As a robustness test, we plot estimates of ν^E for realistic levels of $\hat{\nu} = \nu^{\text{1-period}}/\nu^E$ as coefficients on μ_{t+1}^{nn}: the estimated ν^E are stable and consistent near our preferred estimate.

Figure A.1: Estimated ν^E for given $\hat{\nu}$



A.2 Proofs of Propositions in Section 4

This appendix provides the formal derivations for the lemmas and propositions presented in Section 4.

A.2.1 Proof of Lemma 1

Lemma 1. (Labor supply response to wages) *The marginal response of labor supply to future wages is given by*

$$x_t^s := \frac{\partial L_{t+1}^s}{\partial w_{t+1}^s} = \frac{\beta}{w_{t+1}^s}\left[\frac{\theta^2}{\nu}\sum_n L_t^n \mu_t^{ns}(1-\mu_t^{ns}) + \frac{1}{\nu^E}E_t^s(1-E_t^s)N_t\right] > 0.$$

Proof. The total labor supply in sector s at time $t+1$ is given by the law of motion (Equation 5), replacing the destination index n with s:

$$L_{t+1}^s = \sum_n \mu_t^{ns}\theta L_t^n + E_t^s N_t$$

We differentiate this with respect to V_{t+1}^s, using the standard logit derivative $\frac{\partial \mu_t^{ns}}{\partial V_{t+1}^s} = \frac{\beta\theta}{\nu}\mu_t^{ns}(1-\mu_t^{ns})$ and $\frac{\partial E_t^s}{\partial V_{t+1}^s} = \frac{\beta}{\nu^E}E_t^s(1-E_t^s)$:

$$\begin{aligned}\frac{\partial L_{t+1}^s}{\partial V_{t+1}^s} &= \sum_n \theta L_t^n\left(\frac{\beta\theta}{\nu}\mu_t^{ns}(1-\mu_t^{ns})\right) + N_t\left(\frac{\beta}{\nu^E}E_t^s(1-E_t^s)\right)\\ &= \beta\left[\frac{\theta^2}{\nu}\sum_n L_t^n\mu_t^{ns}(1-\mu_t^{ns}) + \frac{1}{\nu^E}N_tE_t^s(1-E_t^s)\right]\end{aligned}$$

From the value function identity (Equation A.2) and $u(w) = \log w$, the second term is $\frac{\partial V_{t+1}^s}{\partial w_{t+1}^s} = \frac{1}{w_{t+1}^s}$, holding future values fixed. Multiplying the terms gives the result:

$$x_t^s = \frac{\beta}{w_{t+1}^s}\left[\frac{\theta^2}{\nu}\sum_n L_t^n\mu_t^{ns}(1-\mu_t^{ns}) + \frac{1}{\nu^E}N_tE_t^s(1-E_t^s)\right]$$

□

A.2.2 When Does the Entry Margin Increase the Supply Elasticity?

This section derives the condition under which the entry margin increases the aggregate labor supply elasticity x_t^s relative to a model with only incumbents. The comparison is between two models of the same economy: the canonical framework without cohort replacement ($\theta = 1$, $N_t = 0$) and our baseline with entry and exit ($\theta < 1$, $N_t > 0$).

Setup. From Lemma 1, the supply slope is

$$x_t^s = \frac{\beta}{w_{t+1}^s}\left[\frac{\theta^2}{\nu}\sum_n L_t^n\mu_t^{ns}(1-\mu_t^{ns}) + \frac{1}{\nu^E}E_t^s(1-E_t^s)N_t\right].$$

To isolate the role of cohort replacement, suppose the total labor force is held fixed via the demographic rule $N_t = (1-\theta)L_t$, where $L_t = \sum_s L_t^s$. A lower θ means faster turnover: more incumbents retire each period, replaced by an equal mass of entrants. The incumbents-only model corresponds to $\theta = 1$.

Derivation. Substituting $N_t = (1-\theta)L_t$ into x_t^s and differentiating with respect to θ:

$$\frac{dx_t^s}{d\theta} = \frac{\beta}{w_{t+1}^s}\left[\frac{2\theta}{\nu}\sum_n L_t^n \mu_t^{ns}(1-\mu_t^{ns}) - \frac{1}{\nu^E}E_t^s(1-E_t^s)L_t\right].$$

The first term captures the mechanical effect of θ on the incumbent margin (more survivors means more potential switchers), and the second captures the loss of entrants as θ rises. The entry margin increases the supply elasticity—that is, x_t^s is decreasing in θ—whenever $dx_t^s/d\theta \leq 0$, which requires

$$\frac{2\theta}{\nu}\sum_n L_t^n \mu_t^{ns}(1-\mu_t^{ns}) \leq \frac{1}{\nu^E}E_t^s(1-E_t^s)L_t. \tag{A.7}$$

Sufficient condition via Jensen's inequality. Define the employment-weighted average inflow share

$$q_t^s := \frac{1}{L_t}\sum_n L_t^n \mu_t^{ns},$$

which is the fraction of the incumbent pool that flows into sector s next period. Since $g(p) = p(1-p)$ is concave on $[0,1]$, Jensen's inequality implies

$$\frac{1}{L_t}\sum_n L_t^n \mu_t^{ns}(1-\mu_t^{ns}) \leq q_t^s(1-q_t^s).$$

Substituting into (A.7) and rearranging yields the sufficient condition

$$\theta \leq \frac{\nu}{2\nu^E}\frac{E_t^s(1-E_t^s)}{q_t^s(1-q_t^s)}. \tag{A.8}$$

Interpretation. Condition (A.8) is satisfied when three forces align: (i) entrants are more responsive than incumbents ($\nu/\nu^E > 1$), (ii) the turnover rate $1-\theta$ is not too small, and (iii) the entrant choice slack $E_t^s(1-E_t^s)$ is non-negligible relative to the incumbent inflow slack $q_t^s(1-q_t^s)$.

Back-of-the-envelope verification. Using our benchmark estimates $\nu \approx 2.44$, $\nu^E \approx 0.350$, and $\theta = 0.97$, the right-hand side of (A.8) becomes

$$\frac{\nu}{2\nu^E} \frac{E_t^s(1-E_t^s)}{q_t^s(1-q_t^s)} \approx 3.27 \times \frac{E_t^s(1-E_t^s)}{q_t^s(1-q_t^s)}.$$

Over the 1985–2019 sample period, observed entry-to-inflow ratios for manufacturing satisfy $E_t^s/q_t^s \in (0.5, 2)$. In such cases, the ratio of slacks $E_t^s(1-E_t^s)/q_t^s(1-q_t^s)$ remains bounded away from zero throughout this range. In all years, the right-hand side of (A.8) comfortably exceeds $\theta = 0.97$, confirming that the entry margin is an empirically relevant source of supply elasticity throughout the sample.

A.2.3 Proof of Proposition 2

Proof. This is a comparative statics exercise in the equilibrium of period $t+1$. The equilibrium is defined by two equations, (1) Labor Supply and (2) Labor Demand, which jointly determine L_{t+1}^s and w_{t+1}^s. Let $L = L_{t+1}^s$, $w = w_{t+1}^s$, and $\xi = \xi_{t+1}^s$.

$$L = L(w, \cdot) \tag{A.9}$$

$$w = f^s(L, \xi) \tag{A.10}$$

We analyze the system by total differentiation. Let $x_t^s = \frac{\partial L}{\partial w}$ be the labor supply elasticity from Lemma 1. Let $f_L^s = \frac{\partial f^s}{\partial L} < 0$ be the slope of the inverse labor demand, and $f_\xi^s = \frac{\partial f^s}{\partial \xi} > 0$ be the effect of the demand shifter.

Differentiating the system:

$$dL = x_t^s dw$$
$$dw = f_L^s dL + f_\xi^s d\xi$$

Solving this system yields the general equilibrium responses to the demand shock ξ:

$$\frac{dw}{d\xi} = \frac{f_\xi^s}{1 - f_L^s x_t^s} \quad \text{and} \quad \frac{dL}{d\xi} = \frac{x_t^s f_\xi^s}{1 - f_L^s x_t^s}$$

An active, elastic entry margin increases the aggregate labor supply elasticity x_t^s. We analyze the comparative statics with respect to x_t^s.

Proof of 1 (Wage Pass-Through):

$$\frac{dw}{d\xi} = \frac{f_\xi^s}{1 - f_L^s x_t^s}$$

The model with an active, elastic entry margin has a strictly larger x_t^s (as x_t^s is increasing in $1/\nu^E$ and N_t). The derivative of the wage response with respect to x_t^s:

$$\frac{\partial}{\partial x_t^s}\left(\frac{dw}{d\xi}\right) = \frac{\partial}{\partial x_t^s}\left(\frac{f_\xi^s}{1 - f_L^s x_t^s}\right) = f_\xi^s \cdot (-1) \cdot (1 - f_L^s x_t^s)^{-2} \cdot (-f_L^s) = \frac{f_L^s f_\xi^s}{(1 - f_L^s x_t^s)^2} < 0$$

(since $f_L^s < 0, f_\xi^s > 0$). A larger x_t^s reduces the response $\frac{dw}{d\xi}$, thus attenuating the wage decline $|\Delta w|$ from a negative shock $\Delta\xi < 0$.

Proof of 2 (Quantity Response): Substitute the expression for $\frac{dw}{d\xi}$ back into (1′) to solve for $\frac{dL}{d\xi}$:

$$\frac{dL}{d\xi} = \frac{x_t^s f_\xi^s}{1 - f_L^s x_t^s}$$

Taking the derivative, we get

$$\frac{\partial}{\partial x_t^s}\left(\frac{dL}{d\xi}\right) = \frac{f_\xi^s(1 - f_L^s x_t^s) - (x_t^s f_\xi^s)(-f_L^s)}{(1 - f_L^s x_t^s)^2} = \frac{f_\xi^s}{(1 - f_L^s x_t^s)^2} > 0$$

(since $f_\xi^s > 0$). A larger x_t^s increases the response $\frac{dL}{d\xi}$, thus amplifying the employment fall $|\Delta L|$ from a negative shock $\Delta\xi < 0$.

Proof of 3 (Calibration): To match an observed $\Delta L^* < 0$, the required shock is $\Delta\xi \approx \Delta L^*/(dL/d\xi)$. From (2), a larger x_t^s increases the denominator $\frac{dL}{d\xi}$. Therefore, a smaller shock magnitude $|\Delta\xi|$ is required to generate the same ΔL^*. □

A.2.4 Proof of Proposition 3

Proof. From the law of motion (5), the total employment response in sector s to a change in continuation values is

$$\frac{\partial L_{t+1}^s}{\partial V_{t+1}^s} = \sum_n S_{I,n} \frac{\partial \mu_t^{ns}}{\partial V_{t+1}^s} + S_E \frac{\partial E_t^s}{\partial V_{t+1}^s},$$

where $S_{I,n} := \theta L_t^n$ is the surviving incumbent mass from sector n and $S_E := N_t$ is the entrant mass. From the logit choice probabilities, the derivatives are

$$\frac{\partial \mu_t^{ns}}{\partial V_{t+1}^s} = \frac{\beta\theta}{\nu} \mu_t^{ns}(1 - \mu_t^{ns}),$$

$$\frac{\partial E_t^s}{\partial V_{t+1}^s} = \frac{\beta}{\nu^E} E_t^s(1 - E_t^s).$$

Define the elasticity-slack product for each margin: $A_{I,n} := S_{I,n} \cdot (\beta\theta/\nu) \cdot \mu_t^{ns}(1-\mu_t^{ns})$ and $A_E := S_E \cdot (\beta/\nu^E) \cdot E_t^s(1-E_t^s)$. Since any demand shock $\Delta\xi_{t+1}^s$ affects employment only through the equilibrium change in V_{t+1}^s, the share of adjustment attributable to margin m is

$$\text{Share}_m = \frac{A_m}{\sum_n A_{I,n} + A_E}, \tag{A.11}$$

which is proportional to size $\times$ elasticity $\times$ slack.

Part (1): Small shocks. For small ΔV_{t+1}^s, the shares in (A.11) are evaluated at baseline values. Since $\beta/\nu^E \gg \beta\theta/\nu$ (from $\nu^E \ll \nu$), the entrant elasticity-slack product A_E is large relative to $A_{I,n}$ whenever E_t^s is bounded away from 0 and 1 (so that $E_t^s(1-E_t^s)$ is non-negligible) and the entrant mass S_E is positive. In this case, entrants account for a disproportionate share of the total employment adjustment.

Part (2): Large cumulative shocks. As $V_{t+1}^s \to -\infty$, both μ_t^{ns} and E_t^s converge to zero, so the slack terms satisfy $p_m^s(1-p_m^s) \approx p_m^s$ for each margin m. From the logit form, the choice probabilities decay exponentially:

$$\mu_t^{ns} \propto \exp\left(\frac{\beta\theta}{\nu} V_{t+1}^s\right), \qquad E_t^s \propto \exp\left(\frac{\beta}{\nu^E} V_{t+1}^s\right),$$

absorbing cost terms and other continuation values into the proportionality constants. The ratio of entrant to incumbent marginal adjustment is therefore

$$\frac{A_E}{A_{I,n}} \propto \frac{\exp(\beta V_{t+1}^s/\nu^E)}{\exp(\beta\theta V_{t+1}^s/\nu)} = \exp\left(\beta V_{t+1}^s \left[\frac{1}{\nu^E} - \frac{\theta}{\nu}\right]\right).$$

The bracketed term $B := 1/\nu^E - \theta/\nu$ is strictly positive since $\nu^E < \nu$ and $\theta \leq 1$. As $V_{t+1}^s \to -\infty$, the exponent $\beta V_{t+1}^s \cdot B \to -\infty$, so $A_E/A_{I,n} \to 0$. The entrant margin's share of marginal adjustment vanishes, and further adjustment is borne increasingly by the less elastic incumbent stock. □

B Details for Empirics

B.1 Data Construction for the Monthly CPS

CPS microdata and linked panel. We use IPUMS CPS Basic Monthly microdata (Flood et al., 2024) from January 1982 through December 2019 and construct one-year links by matching respondents interviewed in the same calendar month one year apart. We restrict attention to civilian, non-institutionalized men ages 20–64 and compute all moments using CPS survey

weights. The CPS rotation design (months-in-sample 1–4 and 13–16) implies that many individuals can be observed at multiple 12-month intervals, so annual transition matrices are constructed at the person-year level using these within-person links.

Matching procedure and validation rules. We link individuals 12 months apart using the IPUMS CPS person identifiers and month-in-sample structure (same calendar month one year apart). We retain links only when sex, race, quarter of birth, and state match and age advances by one year (allowing standard minor reporting error). Because the CPS rotation pattern can generate multiple valid 12-month links for the same individual, we use a single canonical link, month-in-sample 4 to 16 (MIS 4→16), to construct annual transitions. We drop ambiguous one-to-many matches and any links failing these checks. Weekly earnings are taken from the outgoing rotation groups and are linked to subsequent one-year transitions using the month-in-sample structure.

State definitions and treatment of unemployment. In each month we assign individuals to one of four mutually exclusive states: manufacturing (M), agriculture (A), services (S), and non-employment (N). Sector states are defined using IPUMS's harmonized 1990 industry classification (`IND1990`), which provides a consistent mapping over time; given our coarse sector partition, remaining classification differences across CPS industry recodes are immaterial.

A key classification choice is the treatment of unemployment. We treat unemployed individuals who report a prior industry as *temporarily separated* and keep them attached to their reported sector, rather than coding them as having exited to N. This convention ensures that short unemployment spells during job search do not mechanically inflate extensive-margin flows; instead, sector switching is recorded when an individual moves between employed sectors across years. In the CPS, reported industry is near-universal ($> 95\%$) among the temporary unemployed but rare ($< 5\%$) among those out of the labor force, consistent with unemployment reflecting temporary separation and N capturing non-employment (Section 2).

Flow reconciliation (biproportional fitting / IPF). Linked CPS transitions produce a raw flow matrix that need not be exactly consistent with observed sectoral stocks in adjacent years: row sums may differ from L_t^i because of linkage gaps, and column sums may differ from L_{t+1}^j because the CPS is a repeated cross-section with changing population composition. To obtain an exact accounting decomposition, we reconcile the raw flows using biproportional fitting (iterative proportional fitting, IPF) on an augmented state space that includes an outside state R capturing entry to and exit from the CPS 20–64 universe.

Concretely, for each year t we begin with raw origin–destination counts $\tilde{F}_t^{ij}$ over $i, j \in \{M, A, S, N, R\}$. We then choose adjusted flows F_t^{ij} that (i) preserve the support of the raw

matrix (cells with $\tilde{F}_t^{ij} = 0$ remain zero), and (ii) match observed stocks in non-outside states:

$$\sum_j F_t^{ij} = L_t^i \quad \text{for } i \in \{M, A, S, N\}, \qquad \sum_i F_t^{ij} = L_{t+1}^j \quad \text{for } j \in \{M, A, S, N\}.$$

All margins involving R are left unconstrained and fit whatever is required for the adjusted matrix to satisfy these stock-flow identities. We implement IPF by alternating row and column rescaling of $\tilde{F}_t$ until the constraints are satisfied to numerical tolerance. The same procedure is applied within origin-date age bins when constructing age-specific decompositions.

B.2 Accounting identities

This appendix collects the accounting identities used to decompose changes in manufacturing employment into intensive and extensive margins, and to separate within-age flow changes from between-age compositional changes. All identities hold exactly for the reconciled flow matrix F_t^{ij} described in Appendix B.1.

Levels identity. Let L_t^i denote the number of individuals in state $i \in \{M, A, S, N\}$ at time t, and let F_t^{ij} denote the number transitioning from i to j between t and $t+1$. Augmenting the state space with an outside state R, the change in sectoral employment levels satisfies

$$L_{t+1}^i - L_t^i = \underbrace{\sum_{j \in \{M,A,S\}} \left(F_t^{ji} - F_t^{ij}\right)}_{\text{intensive}} + \underbrace{\left(F_t^{Ni} - F_t^{iN}\right)}_{\text{extensive}} + \underbrace{\left(F_t^{Ri} - F_t^{iR}\right)}_{\text{outside / residual}}. \tag{B.1}$$

Shares identity and the composition effect. Define total employment $L_t \equiv L_t^M + L_t^A + L_t^S$ and sectoral employment shares $s_t^i \equiv L_t^i / L_t$ for $i \in \{M, A, S\}$. Using (B.1) and allowing the employment denominator to change over time, the exact change in employment shares is

$$s_{t+1}^i - s_t^i = \underbrace{\frac{1}{L_{t+1}} \sum_{j \in \{M,A,S\}} \left(F_t^{ji} - F_t^{ij}\right)}_{\text{intensive}} + \underbrace{\frac{1}{L_{t+1}} \left(F_t^{Ni} - F_t^{iN}\right)}_{\text{extensive}} + \underbrace{\frac{1}{L_{t+1}} \left(F_t^{Ri} - F_t^{iR}\right)}_{\text{outside / residual}} + \underbrace{s_t^i \left(\frac{L_t}{L_{t+1}} - 1\right)}_{\text{composition effect}}. \tag{B.2}$$

The final term captures changes in shares arising from movements in the aggregate employment denominator L_t even when sectoral headcounts are unchanged.

Age-specific decomposition and aggregation. For age groups $g \in \{\text{young, prime-age, old}\}$ defined at the origin date t, let $L_t^i(g)$ and $F_t^{ij}(g)$ denote the corresponding age-specific stocks and flows. Identities analogous to (B.1) and (B.2) hold within each g. We aggregate age-specific

shares into the overall share using the exact decomposition

$$s^i_{t+1} - s^i_t = \underbrace{\sum_g \bar{w}_t(g)\left[s^i_{t+1}(g) - s^i_t(g)\right]}_{\text{within-age (flows)}} + \underbrace{\sum_g \bar{s}^i_t(g)\left[w_{t+1}(g) - w_t(g)\right]}_{\text{between-age (composition)}}, \tag{B.3}$$

where $w_t(g) \equiv L_t(g)/L_t$ is the employment weight of age group g, $\bar{w}_t(g) \equiv \frac{1}{2}\big(w_t(g) + w_{t+1}(g)\big)$, and $\bar{s}^i_t(g) \equiv \frac{1}{2}\big(s^i_t(g) + s^i_{t+1}(g)\big)$.

Entry and exit labels. When summarizing the extensive margin by life-cycle stage, we define "exit" as the flow from manufacturing to non-employment among older workers, $F^{MN}_t(\text{old})$, and "entry" as the flow from non-employment to manufacturing among young workers, $F^{NM}_t(\text{young})$. All age groups are defined at the origin date t.

B.3 Robustness for Section 2

BLS "Fact 1." Figure B.1 reproduces Figure 1 using BLS employment totals for manufacturing headcount (and the corresponding aggregate employment denominator) in place of the CPS-linked headcount series. The same divergence is present and, if anything, more pronounced: manufacturing's employment share declines smoothly over the full sample, while manufacturing headcount is relatively stable through the late 1990s and then falls sharply after 2000. This confirms that the share–headcount break motivating Fact 1 is not an artifact of CPS linking.

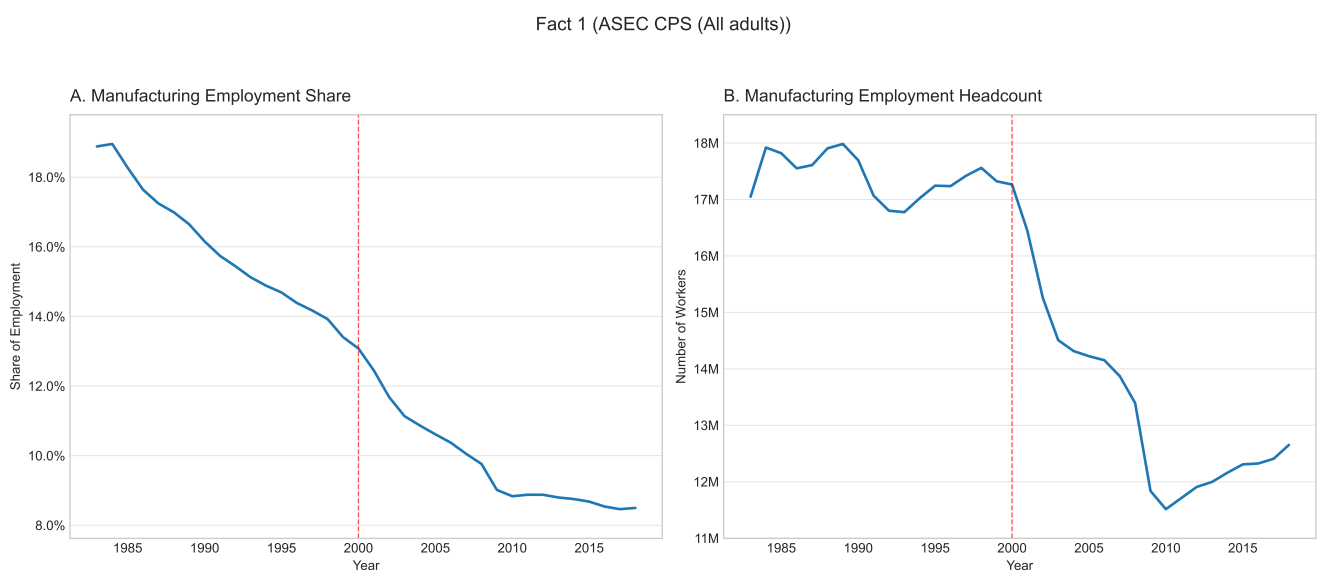


Figure B.1: Divergence of Manufacturing Employment Share and Headcount (BLS).

Using the LEHD dataset.

We conduct robustness exercises for the motivating findings using the Job-to-Job Flows (J2J) data from the Longitudinal Employer-Household Dynamics (LEHD) program.

Data Source and Sample Construction. The J2J dataset from the LEHD program, produced by the U.S. Census Bureau, tracks worker reallocation by linking state Unemployment Insurance (UI) wage records with firm-level data from the Quarterly Census of Employment and Wages (QCEW) and historical demographic records. Achieving near-national coverage by the year 2000, the dataset provides a quarterly panel that covers approximately 95% to 97% of all U.S. wage and salary jobs.

Defining Sectoral Entrants. To measure inflows into a sector sector from nonemployment, we utilize the LEHD J2J NEPersist variable (hires from persistent nonemployment). This variable identifies workers newly hired into a firm who had no reported UI earnings in the preceding quarter. By requiring a full quarter without earnings, NEPersist filters out short-term, frictional spells between jobs and isolates true structural entrants—capturing both school-to-work transitions and workers re-entering the labor force after extended displacement.

Sample Demographics and Sector Classification. Our primary analysis of "young workers" in this section maps to the LEHD age bin of 19–24 years old. To ensure our results are robust and strictly align with the broader CPS age definitions used in the main text, we also conduct a separate exercise incorporating the 25–34 age bin. Across all LEHD exercises, the manufacturing sector is consistently identified using the 2-digit NAICS classification 31–33.

Limitations on Sector-to-Sector Flows. While the J2J data precisely tracks total employment movements in and out of the manufacturing sector (separating hires from nonemployment versus hires from other jobs), a limitation of the data structure used in this exercise is the lack of detailed sector-to-sector transition matrices. Consequently, while we observe the aggregate intensive margin (workers switching into manufacturing from other employed positions), we do not trace the specific originating industries (e.g., Services or Construction) for these incumbent job-to-job switchers.

Results. [Available upon request.]

Using the March CPS.

Data and motivation. To validate that our findings are not an artifact of monthly CPS linking, we replicate the manufacturing decline decomposition using the CPS Annual Social and

Economic Supplement (ASEC, "March CPS"). The ASEC records respondents' industry at the March interview and, separately, the industry in which they spent the most time during the previous calendar year. This provides a retrospective one-year transition measure without relying on month-in-sample matching.

Sample and state construction. We follow the baseline sample restrictions as closely as possible (men ages 20–64; survey weights). We map both current and prior-year industries into the same four-state partition $\{M, A, S, N\}$ using harmonized industry codes. We treat non-employment analogously by coding individuals as N when they report not working, and we construct annual transition matrices from prior-year state to current state.

Results. The ASEC-based decomposition yields the same qualitative conclusion as the linked monthly CPS: the manufacturing decline is dominated by the extensive margin, and age patterns indicate that exits are concentrated among older workers while entry by young workers declines. Appendix Figure A.X and Table A.X report the ASEC analogues of Figures 2–3.

[Results available upon request.]

C Life-cycle dynamics

We are extending the baseline entrant–incumbent framework to a finite-horizon life-cycle model in which mobility costs and retirement risk vary with age (or tenure). This extension nests the baseline model as a parsimonious approximation and clarifies how long-horizon entry decisions and short-horizon late-career decisions jointly shape sectoral adjustment. Details are work in progress and available upon request.